Celebration Set: Proving the Cyclicity of U_p

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Abstract

We prove that U_p, the group of nonzero residue classes modulo a prime p, is cyclic, using modular arithmetic, orders of elements, Euler's φ-function, and a counting argument comparing the number of elements of each possible order. The proof is developed fully from mathematical axioms.

1 Introduction

1.1 Preface

This document began as a collaborative project by Family 41 at the 2026 Ross Mathematics Program, which we completed over a weekend. The original document was about 60 pages long, but after the program, I extensively revised the proofs and layout, added some additional lemmas, and created a dependency diagram (Figure 1) showing dependencies among Section 3.6 and Main Results.

1.2 Proof Outline

We begin by stating the Ring and Order Axioms in Section 2, and then presenting some preliminary definitions and lemmas in Section 3. Finally, our main sequence of results is presented in Section 4. Proofs that are excessively long and not central to our main claim are given in the Appendix.

The proof has four main steps. First, we define U_p and show that it has $p-1$ elements. Next, we show that the order of every element of U_p divides $p-1$. For each positive divisor $d \mid p-1$, let $N(d)$ denote the number of elements of order d. We prove that $N(d) \leq \phi(d)$ using a root-counting argument. Finally, since

$$\sum_{d \mid p-1} N(d) = p-1 \qquad \text{and} \qquad \sum_{d \mid p-1} \phi(d) = p-1,$$

we conclude that $N(p-1) = \phi(p-1) \geq 1$, so U_p contains an element of order $p-1$. Hence U_p is cyclic.

Contents

2 Axioms

2.1 Ring Axioms

Define a set R, along with two binary operations $+$ and $\cdot$, and abbreviate $a \cdot b$ as ab. Then R is a **ring** if it satisfies, for all $a, b, c \in R$,

1. Commutativity: $a + b = b + a$, and $a \cdot b = b \cdot a$.
2. Associativity: $a + (b + c) = (a + b) + c$, and $a(bc) = (ab)c$.
3. Distributivity: $a(b + c) = ab + ac$.
4. Zero: There exists $0 \in R$ such that for all $a \in R$, $a + 0 = a$.
5. Negatives: For all $a \in R$, there exists $x \in R$ such that $a + x = 0$.
6. One: There exists $1 \in R$ such that for all $a \in R$, $a \cdot 1 = a$.

2.2 Order Axioms

Let T be an ordered ring if there exists a nonempty subset $P \subseteq T$ such that

1. Closure: For all $a, b \in P, a + b \in P$ and $a \cdot b \in P$,
2. Non-triviality: $0 \notin P$,
3. Trichotomy: For all $a \in T$, exactly one of the following holds: $a \in P, a = 0, (-a) \in P$ (see Definition 3.3)

We assume that $\mathbb{Z}$ is an ordered ring.

The **Well-Ordering Principle** (often abbreviated as WOP) states that all nonempty subsets of $Z^{\geq 0}$ contain a "least element."

Definition 2.1 (Least Element) — Let $S \subseteq T$, where T is an ordered ring. We say $m \in S$ is a least element of S if, for all $x \in S$, $m \leq x$.

We prove that m is unique in Lemma 3.17.

3 Definitions and Lemmas

Let R be an arbitrary ring and T an arbitrary ordered ring.

3.1 Basic Ring Identities

Lemma 3.1 (Additive Cancellation)**.** For all $a, b, c \in R$,

$$a + b = a + c \implies b = c,$$

$$b + a = c + a \implies b = c.$$

Proof. By the Negatives axiom, there exists $h \in R$ such that $a + h = 0$. Hence, $h + a = 0$ by commutativity. Then,

$$\begin{aligned} h + (a + b) &= h + (a + c), \\ (h + a) + b &= (h + a) + c, \\ 0 + b &= 0 + c, \\ b + 0 &= c + 0, \\ b &= c. \end{aligned}$$

Also, for the second case, $(b + a) + h = (c + a) + h$, so

$$\begin{aligned} b + (a + h) &= c + (a + h), \\ b + 0 &= c + 0, \\ b &= c, \end{aligned}$$

as desired. □

Lemma 3.2 (Uniqueness of Additive Inverse)**.** For all $a \in R$, there exists unique $h \in R$ such that $a + h = 0$.

Proof. Existence follows from the Negatives axiom. Now say that $a + h_1 = a + h_2 = 0$. Then by Lemma 3.1, $h_1 = h_2$. □

Definition 3.3 (Additive Inverse) — We denote the unique additive inverse of $a \in R$ as $-a$, such that $a + (-a) = 0$.

Lemma 3.4 (Identity Identities)**.** For all $a \in R$,

(i) $0 + a = a$.

(ii) $1 \cdot a = a$.

(iii) $a \cdot 0 = 0$.

Proof. (i) $0 + a = a + 0 = a$.

(ii) $1 \cdot a = a \cdot 1 = a$.

(iii) Note that

$$\begin{aligned} a \cdot 0 + a \cdot 0 &= a(0+0) \\ &= a \cdot 0 \\ &= a \cdot 0 + 0, \end{aligned}$$

and hence, by Lemma 3.1, $a \cdot 0 = 0$.

□

Lemma 3.5 (Existence and Uniqueness of Subtraction)**.** For all $a, b \in R$, there exists a unique $y \in R$ such that

$$a = y + b.$$

Proof. (Existence) Let $a, b \in R$. By the ring axioms, there exists $-b \in R$ such that $b + (-b) = 0$. Let $y = a + (-b)$. Then

$$y + b = (a + (-b)) + (b) = a + ((-b) + b) = a + 0 = a.$$

Thus, a solution to $a = y + b$ exists. □

Proof. (Uniqueness) Say $y_1, y_2 \in R$ satisfy $a = y_1 + b$ and $a = y_2 + b$. Then $y_1 + b = y_2 + b$, so $y_1 = y_2$ by Lemma 3.1. Thus the solution y is unique. □

Definition 3.6 (Subtraction) — For $a, b \in R$, let $a - b$ be the unique solution $y \in R$ to the equation $a = y + b$.

Lemma 3.7 (Explicit Formula for Subtraction)**.** For all $a, b \in R$, $a - b = a + (-b)$.

Proof. We know that $a = (a - b) + b$ by definition. But since

$$(a + (-b)) + b = a + ((-b) + b) = a + 0 = a,$$

we have $a - b = a + (-b)$ by Lemma 3.1. □

Lemma 3.8 (Additive Inverse Identities)**.** For all $a, b \in R$,

(i) $(-a) + a = 0$.

(ii) $-(-a) = a$.

(iii) $-a = a \cdot (-1) = (-1) \cdot a$.

(iv) $(-1)(-1) = 1$.

(v) $-(a + b) = (-a) + (-b)$.

(vi) $b - a = (-1)(a - b) = (a - b)(-1)$.

(vii) $a - (a - b) = b$.

(viii) $-(ab) = a(-b) = (-a)b$.

(ix) $(-a)(-b) = ab$.

Proof. The proof is a direct verification from the ring axioms; see Appendix A.1. □

Lemma 3.9 (Adding Equalities). If $a, b, c, d \in R$ satisfy $a = b$ and $c = d$, then $a + c = b + d$.

Proof. We have $a + c = b + c$, and by substitution, $a + c = b + d$. □

Lemma 3.10 (Rearranging Four Terms). If $a, b, c, d \in R$, and $*$ is an associative and commutative binary operator in R (e.g. $+$ or $\cdot$), then

$$(a * b) * (c * d) = (a * c) * (b * d) = (a * d) * (b * c).$$

Proof. We have

$$\begin{aligned}(a * b) * (c * d) &= a * (b * (c * d)) \\ &= a * ((c * d) * b) \\ &= a * (c * (d * b)) \\ &= (a * c) * (d * b) \\ &= (a * c) * (b * d).\end{aligned}$$

Also, from $(a * b) * (c * d) = a * (c * (d * b))$, we have

$$\begin{aligned}(a * b) * (c * d) &= a * ((d * b) * c) \\ &= a * (d * (b * c)) \\ &= (a * d) * (b * c),\end{aligned}$$

as desired. □

3.2 Ordered Ring Identities

Definition 3.11 (Strict Order) — For $a, b \in T$, define $a < b$ if there exists $k \in P$ such that $a + k = b$. We say $a > b$ iff $b < a$.

Definition 3.12 (Non-strict Order) — For $a, b \in T$ define $a \leq b$ if either $a = b$ or $a < b$. We say $a \geq b$ iff $b \leq a$.

Lemma 3.13 (Positive Cone Characterization). For $x \in T$, we have $x \in P$ if and only if $x > 0$.

Proof. First assume $x > 0$. Then by definition, there exists $k \in P$ such that $0 + k = x$, so $k + 0 = x$ and so $k = x$. Thus $x \in P$ by substitution.

Next assume $x \in P$. Then $x + 0 = x$, so $0 + x = x$. Since $x \in P, 0 < x$, so $x > 0$. □

Definition 3.14 (Positive Integers) — Define $\mathbb{Z}^+ = \{x \in \mathbb{Z} : x > 0\}$.

Definition 3.15 (Nonnegative Integers) — Define $\mathbb{Z}^{\geq 0} = \{x \in \mathbb{Z} : x \geq 0\}$.

Lemma 3.16 (Basic Properties of Order)**.** For all $x, y, z \in T$, the following hold:

(i) Exactly one of $x < y$, $x = y$, or $x > y$ is true.

(ii) If $x \geq y$ and $y \geq x$, then $x = y$.

(iii) If $x < y$ and $y < z$, then $x < z$.

(iv) If $x \leq y$ and $y \leq z$, then $x \leq z$.

(v) If $x < y$, then $x + z < y + z$.

(vi) If $z > 0$ and $x < y$, then $xz < yz$.

(vii) If $z > 0$ and $x \leq y$, then $xz \leq yz$.

(viii) If $x < 0$, then $-x > 0$. Similarly, if $x > 0$, then $-x < 0$.

Proof. See Appendix A.2. □

Lemma 3.17 (Uniqueness of Least Element)**.** If m is a least element of $S \subseteq T$, m is unique.

Proof. Say $m_1, m_2 \in S$ both satisfy, for all $x \in S$, $m_1 \leq x$ and $m_2 \leq x$. Then since $m_1 \in S$,

$$m_2 \leq m_1.$$

Also, since $m_2 \in S$,

$$m_1 \leq m_2.$$

Therefore by Lemma 3.16, $m_1 = m_2$. □

Lemma 3.18 (Discreteness of $\mathbb{Z}$)**.** The following hold:

(i) There is no $a \in \mathbb{Z}$ such that $0 < a < 1$.

(ii) If $a, m \in \mathbb{Z}$ and $a > m$, then $a \geq m + 1$.

Proof. See Appendix A.3. □

Lemma 3.19 (Zero Product Property)**.** There exists no $a, b \in T$, $a, b \neq 0$ such that $ab = 0$.

Proof. There are three cases to consider.

(i) $a, b > 0$. If $a, b > 0$, then $a, b \in P$, meaning that by closure, $ab \in P$, and by Trichotomy $ab \neq 0$.

(ii) Without loss of generality, assume $a > 0$ and $b < 0$. Then $b \notin P$ by Lemma 3.13, and as $b \neq 0$ we have $-b > 0$ by Lemma 3.16. Thus by Lemma 3.16,

$$a(-b) > 0.$$

By Lemma 3.8, $a(-b) = -(ab)$. Hence $-(ab) > 0$, and so $-(-(ab)) < 0$. By Lemma 3.8, $ab < 0$, so $ab \neq 0$.

(iii) $a, b < 0$. Then, as above, $-a, -b > 0$. Thus,

$$\begin{aligned}(-a)(-b) &= \big[(-1)a\big]\big[(-1)b\big]\\ &= \big[(-1)(-1)\big](ab)\\ &= 1(ab)\\ &= ab.\end{aligned}$$

Therefore, $ab > 0$, so $ab \neq 0$ by Lemma 3.16.

In all cases, $ab \neq 0$. □

Lemma 3.20 (Multiplicative Cancellation)**.** For all $a, b, c \in T, a \neq 0$, $ab = ac$ implies that $b = c$.

Proof. Since, $ab = ac$, we have

$$\begin{aligned}ab + (-(ac)) &= 0\\ ab + a(-c) &= 0\\ a(b + (-c)) &= 0.\end{aligned}$$

Since $a \neq 0$, by Lemma 3.19, we know $b + (-c) = 0$. But $c + (-c) = 0$, so by Lemma 3.1, $b = c$. □

Lemma 3.21 (Positive Quotient Property)**.** For all $a, b, c \in T$, if $ab = c$ and $a, c > 0$, then $b > 0$.

Proof. Note that $c > 0$, so by Lemma 3.13, $c \in P$. Hence $ab \in P$. Assume for the sake of contradiction that $b < 0$ or $b = 0$. If $b = 0$, then $c = a \cdot b = a \cdot 0 = 0$, contradicting $c > 0$. However, if $b < 0$, then $-b > 0$ by Lemma 3.16, so $a(-b) > 0$ and thus $-(ab) > 0$ by Lemma 3.8. Hence $-(ab) \in P$ by Lemma 3.13, but by Trichotomy, this contradicts $ab \in P$. Thus by contradiction, $b > 0$. □

3.3 Divisibility

Definition 3.22 (Divisibility) — For $a, b \in R$, $a \mid b$ if $b = ak$ for some $k \in R$.

Lemma 3.23 (Uniqueness of Quotient)**.** Let $d, a \in \mathbb{Z}$ with $d \neq 0$ and $d \mid a$. If $p, q \in \mathbb{Z}$ satisfy $dp = a$ and $dq = a$, then $p = q$.

Proof. (Uniqueness) Suppose $p, q \in \mathbb{Z}$ satisfy $dp = a$ and $dq = a$. By substitution, $dp = dq$, so by Lemma 3.20, we have $p = q$. Hence integer division is unique. □

Definition 3.24 (Division) — For $d, a \in \mathbb{Z}$, $d \neq 0$ and $d \mid a$, define the unique solution $dk = a$, $k \in \mathbb{Z}$ as the quotient, denoted as $k = \frac{a}{d}$.

Lemma 3.25 (Cancelling a Quotient). Let $a, b \in \mathbb{Z}$ with $b \neq 0$, $b \mid a$, and $\frac{a}{b} \neq 0$. Then

$$\frac{a}{a/b} = b.$$

Proof. Let $k = \frac{a}{b}$, then by definition $bk = a$. Then, let $q = \frac{a}{k}$, so $qk = a$. Then by substitution, $bk = qk$, and by 3.20, this implies that $b = q$. Hence, $\frac{a}{\frac{a}{b}} = b$. □

Lemma 3.26 (Linear Combination Divisibility). If $a \mid b$ and $a \mid c$, then for any $x, y \in \mathbb{Z}$, $a \mid (bx + cy)$.

Proof. See Appendix B.1. □

Lemma 3.27 (Divisibility Ordering). For $a, b \in \mathbb{Z}^+$, if $a \mid b$, then $a \leq b$.

Proof. $a \mid b$ implies there exists a $c \in \mathbb{Z}$ such that $ac = b$. Since $a > 0$ and $b > 0$, Lemma 3.21 gives $c > 0$. Then by Lemma 3.18, $c \geq 1$. Since $a > 0$, Lemma 3.16 gives

$$a \cdot 1 \leq ac.$$

Therefore $a \leq b$. □

Definition 3.28 (Prime) — Let $p \in \mathbb{Z}^+$ be prime if $p > 1$, and for every $a, b \in \mathbb{Z}$, if $p \mid ab$, then $p \mid a$ or $p \mid b$.

Definition 3.29 (Irreducible) — Let $q \in \mathbb{Z}^+$. We say q is irreducible if $q > 1$ and there do not exist $a, b \in \mathbb{Z}^+$ with $a \geq 2, \quad b \geq 2$, such that $ab = q$.

Lemma 3.30 (Primes Are Irreducible). Let p be prime, then p is irreducible.

Proof. Suppose p is prime, and assume for the sake of contradiction that p is not irreducible. This means that for some $a, b \in \mathbb{Z}^+$ with $a, b \geq 2$, $p = ab$. Because $p = p \cdot 1 = ab$ and $1 \in \mathbb{Z}$, we have $p \mid ab$. Then by definition, $p \mid a$ or $p \mid b$. If $p \mid a$, then by Lemma 3.27, $p \leq a$. Since $b \geq 2$, we know $b > 1$. Then Lemma 3.16 gives $p < bp$ and $bp \leq ba$. Hence, $p < ab$, contradicting $p = ab$. The contradiction when $p \mid b$ follows similarly, so by contradiction, p must be irreducible. □

Lemma 3.31 (Every Element Divides Zero). For all $m \in R$, $m \mid 0$.

Proof. By Lemma 3.4, $m \cdot 0 = 0$. Since $0 \in R$, $m \mid 0$. □

3.4 Exponents

Definition 3.32 (Exponents) — Let R be a ring. For $a \in R$ and $b \in \mathbb{Z}^{\geq 0}$, define:

(i) $a^0 = 1$,

(ii) for all $b \in \mathbb{Z}^+$, $a^b = (a^{b-1}) \cdot a$.

Lemma 3.33 (Exponent Properties). Let $a \in R$. Then for all $m, n \in \mathbb{Z}^+$,

(i) $a^{m+n} = a^m \cdot a^n$.

(ii) $a^{mn} = (a^m)^n$.

Proof. See Appendix B.2. ☐

3.5 Congruence Classes and Modular Arithmetic

Lemma 3.34 (Division Algorithm). For every $a, b \in \mathbb{Z}$ where $b \in P$, there exists integers q, r where $0 \leq r < b$ such that $a = bq + r$.

Proof. Let

$$S = \{a - bq \in \mathbb{Z}^{\geq 0} \mid q \in \mathbb{Z}\}.$$

We claim that S is nonempty. If $a \geq 0$, let $q = 0$. So $a - bq = a - 0 = a$, thus $a \in S$. If $a < 0$, let $q = a$. So $a - bq = a - ba = a(1 - b) = -a(b - 1)$. By 3.18, $b \geq 1$, thus $b - 1 \geq 0$. Since $a < 0$, we have $-a > 0$ by Lemma 3.16, so by Lemma 3.16, $-a(b - 1) \geq 0$. Hence $a - ba \in S$, so either way, S is nonempty.

Hence by WOP, S has a least element $r \in \mathbb{Z}^{\geq 0}$, where for some $q \in \mathbb{Z}$, $r = a - bq$.

We know that $r \geq 0$, so it remains to show that $r < b$. By Lemma 3.16, exactly one of $r < b$, $r \geq b$ is true. Assume for the sake of contradiction that $r \geq b$. Then $r - b \geq 0$.

Substituting,

$$r - b = (a - bq) - b = a - b(q + 1).$$

Since $q + 1 \in \mathbb{Z}$, $r - b \in S$. But $b \in P$, so since $(r - b) + b = r$, we have $r - b < r$. Then, $r - b \in S$, contradicting the minimality of r. Hence by contradiction, $r < b$.

Therefore, there exist $q, r \in \mathbb{Z}$ such that $a = bq + r$ and $0 \leq r < b$. ☐

Definition 3.35 (GCD) — Let $a, b \in \mathbb{Z}$ with $(a, b) \neq (0, 0)$. We say $d \in \mathbb{Z}$ is the greatest common divisor of a and b, denoted

$$d = \gcd(a, b),$$

if

(i) $d \mid a$ and $d \mid b$,

(ii) for every $c \in \mathbb{Z}$, if $c \mid a$ and $c \mid b$, then $c \mid d$,

(iii) $d \in \mathbb{Z}^+$.

Lemma 3.36 (Existence and Uniqueness of GCD). For all $a, b \in \mathbb{Z}$ with $(a, b) \neq (0, 0)$, the greatest common divisor $\gcd(a, b)$ exists and is unique.

Proof. (Existence) Let

$$S = \{m \in \mathbb{Z}^+ : m = ax + by, x, y \in \mathbb{Z}\}.$$

This set is nonempty because:

- When $a > 0$: If $x = 1, y = 0$, $a \cdot 1 + b \cdot 0 = a \cdot 1 + 0 = a \cdot 1 = a$. Since $a \in \mathbb{Z}^+$, $a \in S$.
- When $a < 0$: If $x = -1, y = 0$, $a(-1) + b(0) = -a$. Since $-a > 0$ by Lemma 3.16, $-a \in S$.
- When $a = 0$: Then $b \neq 0$. Applying the steps above to b we get $b \in S$ if $b > 0$ and $-b \in S$ if $b < 0$.

Thus, by WOP, S has a least element d. Let $d = ax_0 + by_0$ for $x_0, y_0 \in \mathbb{Z}$.
By Lemma 3.34, $a = qd + r$ for $0 \leq r < d$. Hence $r = a - qd$, so

$$\begin{aligned} r &= a - q(ax_0 + by_0) \\ &= a - (q(ax_0) + q(by_0)) \\ &= (a - q(ax_0)) - q(by_0) \\ &= (a - a(qx_0)) - q(by_0) \\ &= a(1 - qx_0) - q(by_0) \\ &= a(1 - qx_0) + b(-qy_0). \end{aligned}$$

Because r can be represented as a linear combination of a and b and $r < d$, $r = 0$, because otherwise, we would have $r \in S$, contradicting the minimality of d in S.

Thus, $a = qd + 0 = qd$ which implies that $d \mid a$. By identical reasoning, $d \mid b$.

We must now show that for any $c \in \mathbb{Z}$ where $c \mid a$ and $c \mid b$, we have $c \mid d$. Because $c \mid a$ and $c \mid b$, c must divide any linear combination of a and b by Lemma 3.26. Therefore, $c \mid ax_0 + by_0$, and $c \mid d$.

Thus d satisfies all three properties in the definition. Therefore

$$d = \gcd(a, b),$$

so the greatest common divisor exists. □

Proof. (Uniqueness) Say we have for $d, d' \in \mathbb{Z}^+$, $d = \gcd(a, b)$ and $d' = \gcd(a, b)$. By the definition of gcd, we have by Definition 3.35(i), $d \mid a$ and $d \mid b$, as well as $d' \mid a$, $d' \mid b$. Hence $d \mid d'$ and $d' \mid d$ by Definition 3.35(ii).

By Lemma 3.27, $d \mid d'$ implies $d \leq d'$. Likewise, $d' \mid d$ implies $d' \leq d$. Hence Lemma 3.16 gives $d = d'$. □

Definition 3.37 (Equivalence Relation) — A relation $\sim$ on a set S is an equivalence relation if, for all $a, b, c \in S$,

(i) $a \sim a$,

(ii) if $a \sim b$, then $b \sim a$,

(iii) if $a \sim b$ and $b \sim c$, then $a \sim c$.

Definition 3.38 (Congruence Modulo m) — Let $m \in \mathbb{Z}$ with $m \neq 0$. For $a, b \in \mathbb{Z}$, we say

$$a \equiv b \pmod m$$

iff

$$m \mid a - b.$$

Lemma 3.39 (Bezout's Identity)**.** For all $a, b \in \mathbb{Z}$ with $(a, b) \neq (0, 0)$, there exist $x, y \in \mathbb{Z}$ such that

$$ax + by = \gcd(a, b).$$

Proof. See Appendix B.3. ☐

Lemma 3.40 (Congruence Is an Equivalence Relation)**.** Let $m \in \mathbb{Z}$ with $m \neq 0$. Congruence modulo m is an equivalence relation on $\mathbb{Z}$.

Proof. See Appendix B.4. ☐

Definition 3.41 (Equivalence Class) — Let $\sim$ be an equivalence relation on a set S. For $a \in S$, define

$$[a] = \{x \in S : x \sim a\}.$$

Lemma 3.42 (Equivalence Class Equality)**.** Let $\sim$ be an equivalence relation on a set S. For $a, b \in S$, $[a] = [b]$ if and only if $a \sim b$.

Proof. First assume $[a] = [b]$. Then for all $x \in [a]$, $x \sim a$. But since $[a] = [b]$, $x \in [b]$. Thus $x \sim b$. By symmetry, $a \sim x$, and by transitivity, $a \sim b$.

Next assume for some $a, b \in S$, we have $a \sim b$. Then let $x \in [a]$ be arbitrary. Then $x \sim a$. But since $a \sim b$, we have $x \sim b$ by transitivity. Thus $x \in [b]$. Similarly, if we take arbitrary $y \in [b]$, then $y \sim b$. But by symmetry, $b \sim a$, so $y \sim a$. Thus $y \in [a]$. Then $[a] \subseteq [b]$ and $[b] \subseteq [a]$, so by set theory, $[a] = [b]$. ☐

Definition 3.43 (Integers Modulo p) — Let p be a prime integer. Define

$$\mathbb{Z}_p = \{[a] : a \in \mathbb{Z}\}$$

, where

$$[a] = \{k \in \mathbb{Z} : k \equiv a \pmod p\}.$$

Lemma 3.44 (Well-Definedness of Equivalence Class Addition)**.** The definition

$$[a] + [b] = [a + b]$$

for $[a], [b] \in \mathbb{Z}_p$ is well-defined.

Proof. See Appendix B.5. ☐

Definition 3.45 (Equivalence Class Addition) — Define $[a] + [b] = [a + b]$ for $[a], [b] \in \mathbb{Z}_p$.

Lemma 3.46 (Well-Definedness of Equivalence Class Multiplication). The definition

$$[a][b] = [ab]$$

for $[a], [b] \in \mathbb{Z}_p$ is well-defined.

Proof. See Appendix B.6. □

Definition 3.47 (Equivalence Class Multiplication) — Define $[a][b] = [ab]$ for $[a], [b] \in \mathbb{Z}_p$.

Lemma 3.48 ($\mathbb{Z}_p$ Is a Ring). $\mathbb{Z}_p$ is a ring.

Proof. See Appendix B.7. □

Definition 3.49 (Multiplicative Inverse) — Let $u \in R$. We say u has a **multiplicative inverse** $v \in R$ if $uv = 1$.

Definition 3.50 (Unit) — Let $u \in R$. We say u is a **unit** in R if u has a multiplicative inverse in R.

Definition 3.51 (The Unit Group U_p) — Define $U_p = \{u \in \mathbb{Z}_p : \exists v \in \mathbb{Z}_p \text{ such that } uv = [1]\}$.

Lemma 3.52 (Multiplicative Closure of U_p). U_p is closed under multiplication.

Proof. Let $a, b \in U_p$. By definition, there exist $a^{-1}, b^{-1} \in \mathbb{Z}_p$ such that

$$aa^{-1} = [1], \qquad bb^{-1} = [1].$$

By Lemma 3.48, $\mathbb{Z}_p$ is a ring so,

$$(ab)(a^{-1}b^{-1}) = (aa^{-1})(bb^{-1}) = [1][1] = [1].$$

Thus ab has a multiplicative inverse $a^{-1}b^{-1} \in \mathbb{Z}_p$, so $ab \in U_p$. Therefore U_p is closed under multiplication. □

Lemma 3.53 (Zero Product Property in $\mathbb{Z}_p$). Let p be prime. If $[a][b] = [0]$ in $\mathbb{Z}_p$, then $[a] = [0]$ or $[b] = [0]$.

Proof. From $[a][b] = [0]$ we have

$$[ab] = [0],$$

so $p \mid ab - 0$ and hence

$$p \mid ab.$$

Then, since p is prime, $p \mid a$ or $p \mid b$.

Equivalently, $p \mid a - 0$ or $p \mid b - 0$. Hence by Lemma 3.42, $[a] = [0]$ or $[b] = [0]$. □

Definition 3.54 (Order) — Let R be a ring and let $u \in R$ be a unit. We say $\operatorname{ord}(u) = n$ for $n \in \mathbb{Z}^+$ if

(i) $u^n = 1$,

(ii) for every $m \in \mathbb{Z}^+$, if $u^m = 1$, then $n \leq m$.

Lemma 3.55 (Uniqueness of Order)**.** Let R be a ring, and let $u \in R$ be a unit. If $\operatorname{ord}(u)$ exists, then it is unique.

Proof. (Uniqueness) Say $u^a = 1$ and $u^b = 1$ and for every $m \in \mathbb{Z}^+$, if $u^m = 1$, then $a \leq m$ and $b \leq m$. Then a and b are both orders of u. But since $u^a = 1$, $b \leq a$, and since $u^b = 1$, $a \leq b$. Thus $a = b$, so the order of u is unique. □

3.6 Sequences and Functions

Definition 3.56 (Sequence) — Let S be a set. We say a sequence in S is a function

$$a : \mathbb{Z}^{\geq 0} \to S.$$

For $i \in \mathbb{Z}^{\geq 0}$, let

$$a(i) = a_i,$$

and denote the sequence as $(a_i)_{i \in \mathbb{Z}^{\geq 0}}$

Definition 3.57 (Summation Notation) — For $a, b \in \mathbb{Z}^{\geq 0}$ with $b \geq a$, define $\sum$ for sequence $(c_i)_{i \in \mathbb{Z}^{\geq 0}}$:

(i) $\sum_{i=a}^{a} c_i = c_a$,

(ii) If $b > a$, $\sum_{i=a}^{b} c_i = \left(\sum_{i=a}^{b-1} c_i\right) + c_b$

Definition 3.58 (Sum over Divisors) — Let $n \in \mathbb{Z}^+$, and let $(a_i)_{i \in \mathbb{Z}^{\geq 0}}$ be a sequence. Then

$$\sum_{d \mid n} a_d = \sum_{i=1}^{n} f(i) a_i,$$

where

$$f(i) = \begin{cases} 0, & i \nmid n \\ 1, & i \mid n. \end{cases}$$

Definition 3.59 (Product Notation) — For $a, b \in \mathbb{Z}^{\geq 0}$ with $b \geq a$, define $\prod$ for sequence $(c_i)_{i \in \mathbb{Z}^{\geq 0}}$:

(i) $\prod_{i=a}^{a} c_i = c_a$,

(ii) If $b > a$, $\prod_{i=a}^{b} c_i = \left(\prod_{i=a}^{b-1} c_i\right) \cdot c_b$

Definition 3.60 (Finite Index Set) — For $n \in \mathbb{Z}^{\geq 0}$, define

$$[n] = \{i \in \mathbb{Z} : 0 \leq i \leq n-1\}$$

and in particular, $[0] = \emptyset$.

Definition 3.61 (Integer Intervals) — For integers a, b with $a \leq b$, define

$$\mathbb{Z}_{a,b} = \{i \in \mathbb{Z} : a \leq i \leq b\}.$$

Lemma 3.62 (Bijection Composition). Let A, B, C be sets. If $f : A \to B$ and $g : B \to C$ are bijections. Then $h : A \to C$ defined by $h(x) = g(f(x))$, denoted $g \circ f$, is also a bijection.

Proof. See Appendix C.5. □

Lemma 3.63 (Bijection Restriction). Let A, B be sets. If $f : A \to B$ is a bijection and $f(x) = y$ for some $x \in A$ and $y \in B$, then the restriction

$$f' : A \setminus \{x\} \to B \setminus \{y\}, \qquad f'(a) = f(a),$$

is also a bijection.

Proof. (Well-Defined) Let $a \in A \setminus \{x\}$. If $f(a) = y$, then

$$f(a) = f(x),$$

so by injectivity of f, $a = x$, a contradiction. Thus $f(a) \in B \setminus \{y\}$.

(Injective) Let $x_1, x_2 \in A \setminus \{x\}$ be arbitrary such that $f'(x_1) = f'(x_2)$. Then

$$f(x_1) = f(x_2),$$

so by injectivity of f, $x_1 = x_2$.

(Surjective) Let $b \in B \setminus \{y\}$ be arbitrary. Since f is surjective, there exists $a \in A$ such that $f(a) = b$. But $a \neq x$, since otherwise,

$$b = f(a) = f(x) = y,$$

contradicting $b \in B \setminus \{y\}$. Thus $a \in A \setminus \{x\}$, and

$$f'(a) = f(a) = b.$$

Thus f' is bijective. □

Lemma 3.64 (General Commutativity of Sums). Let $n \in \mathbb{Z}^{\geq 0}$. For a sequence $(a_i)_{i \in \mathbb{Z}^{\geq 0}}$, let $f : [n+1] \to [n+1]$ be a bijection. Then

$$\sum_{i=0}^{n} a_i = \sum_{i=0}^{n} a_{f(i)}.$$

Proof. See Appendix C.6. □

Lemma 3.65 (General Product Commutativity)**.** Let $n \in \mathbb{Z}^+$. For a sequence $(a_i)_{i\in\mathbb{Z}^{\geq 0}}$, let $f : [n] \to [n]$ be a bijection. Then
$$\prod_{i=0}^{n-1} a_i = \prod_{i=0}^{n-1} a_{f(i)}.$$

Proof. See Appendix C.7. □

Lemma 3.66 (One-Indexed Product Commutativity)**.** Let $n \in \mathbb{Z}^+$. For a sequence $(a_i)_{i\in\mathbb{Z}^{\geq 0}}$, let $f : \mathbb{Z}_{1,n} \to \mathbb{Z}_{1,n}$ be a bijection. Then
$$\prod_{i=1}^{n} a_i = \prod_{i=1}^{n} a_{f(i)}.$$

Proof. See Appendix C.8. □

Lemma 3.67 (Product Constant Extraction)**.** For $n \in \mathbb{Z}^+$, and a an arbitrary constant from a ring R:
$$\prod_{i=1}^{n} ai = a^n \prod_{i=1}^{n} i$$

Proof. See Appendix C.9. □

Lemma 3.68 (Linearity of Finite Sums)**.** Let $(x_i)_{i\in\mathbb{Z}^{\geq 0}}$ and $(y_i)_{i\in\mathbb{Z}^{\geq 0}}$ be sequences in a ring R. Then, for all $n \in \mathbb{Z}^{\geq 0}$,
$$\sum_{i=0}^{n}(x_i + y_i) = \sum_{i=0}^{n} x_i + \sum_{i=0}^{n} y_i.$$

Proof. See Appendix C.10. □

Definition 3.69 (Finite Set) — A set S is finite if for some $n \in \mathbb{Z}^{\geq 0}$, there exists a bijection $f : [n] \to S$.

Lemma 3.70 (Uniqueness of Finite Cardinality)**.** For $n, m \in \mathbb{Z}^{\geq 0}$, If there exists a bijection
$$f : [n] \to [m],$$
then $m = n$.

Proof. See Appendix C.11. □

Lemma 3.71 (Inverse of a Bijection)**.** Let A and B be sets, and let $f : A \to B$ be a bijection. Define
$$f^{-1} = \{(b, a) \in B \times A : (a, b) \in f\}.$$
Then $f^{-1} : B \to A$ is a bijection.

Proof. See Appendix C.12. □

Lemma 3.72 (Well-Definedness of Cardinality)**.** Let S be a finite set. If there exist bijections

$$f : [n] \to S \qquad \text{and} \qquad g : [m] \to S,$$

then $n = m$. Therefore, $|S|$ is well-defined.

Proof. (Uniqueness) Assume there exist bijections

$$f : [n] \to S, g : [m] \to S.$$

Since g is a bijection, its inverse

$$g^{-1} : S \to [m]$$

is also a bijection by Lemma 3.71. Therefore,

$$g^{-1} \circ f : [n] \to [m]$$

is a bijection by Lemma 3.62.

By Lemma 3.70, we have

$$n = m.$$

Therefore, the cardinality of S is unique. □

Definition 3.73 (Cardinality) — Let S be a finite set. If for some $n \in \mathbb{Z}^{\geq 0}$, there exists a bijection $f : [n] \to S$. then we define $|S| = n$.

Lemma 3.74 (Equal Size)**.** If A, B are finite sets and there exists a bijection $h : A \to B$, then $|A| = |B|$.

Proof. Let $|A| = m$ and $|B| = n$, so that there exist bijections

$$f : [m] \to A, g : [n] \to B.$$

Then since g is a bijection, by Lemma 3.71,

$$g^{-1} : B \to [n]$$

exists and is also a bijection. Also, by Lemma 3.62,

$$h \circ f : [m] \to B$$

is a bijection, and so is

$$g^{-1} \circ (h \circ f) : [m] \to [n].$$

Then by Lemma 3.70, $m = n$, so

$$|A| = |B|.$$

□

Lemma 3.75 (Cardinality of a Disjoint Union)**.** Let A and B be finite sets such that $A \cap B = \emptyset$. Then $|A \cup B| = |A| + |B|$.

Proof. See Appendix C.13. □

Lemma 3.76 (Subsets of Finite Sets Are Finite)**.** Let B be a finite set and let $A \subseteq B$. Then A is finite.

Proof. See Appendix C.14. $\square$

Lemma 3.77 (Equal-Size Subsets Are Equal)**.** Let A and B be finite sets. If $A \subseteq B$ and $|A| = |B|$, then $A = B$.

Proof. Let $S = \{n \in \mathbb{Z}^{\geq 0} : \exists A, B, A \subseteq B, A \neq B, |A| = |B| = n\}$. Assume for contradiction that S is nonempty. Then by WOP, it has least element n.

We claim that $n \neq 0$. Assume for contradiction that $n = 0$. Then if $|A| = |B| = 0$, $A = \emptyset$ and $B = \emptyset$. Thus $A = B$, a contradiction.

Thus $n > 0$, so $n \geq 1$ by 3.18. Thus $n - 1 \geq 0$, but $n - 1 < n$, so by minimality of $n \in S$, we have $n - 1 \notin S$. Thus for all A, B such that $A \subseteq B$ and $|A| = |B| = n - 1$, we have $A = B$.

Now choose sets A, B such that $A \subseteq B, |A| = |B| = n$, but $A \neq B$. Then we may choose some $a \in A$ because $|A| = n \geq 1$. Then $a \in B$, so

$$A \setminus \{a\} \subseteq B \setminus \{a\}.$$

By Lemma 3.76, $A \setminus \{a\} \subset A$ and $B \setminus \{a\} \subset B$ are finite. Also, $\{a\}$ is finite and $|\{a\}| = 1$, since we can verify that

$$f : [1] \to \{a\}, \qquad f(0) = a$$

is a bijection.

Also, by set theory,

$$A = (A \setminus \{a\}) \cup \{a\}, \qquad (A \setminus \{a\}) \cap \{a\} = \emptyset.$$

Thus we can apply Lemma 3.75 to obtain

$$|A| = |A \setminus \{a\}| + |\{a\}|,$$

so

$$|A \setminus \{a\}| = n - 1.$$

Similarly,

$$|B \setminus \{a\}| = n - 1.$$

Thus since $n - 1 \notin S$,

$$A \setminus \{a\} = B \setminus \{a\}.$$

But taking the union of $\{a\}$ with both sides, this means $A = B$, contradicting $n \in S$. Thus S is empty; the identity is true for all sizes of finite sets. $\square$

Definition 3.78 (Partition) — Let S be a set and I an index set. A set of sets $\{A_i\}_{i \in I}$ is a **partition** of S iff

(i) $S = \bigcup_{i \in I} A_i$,

(ii) For all $i, j \in I$, if $i \neq j$, then $A_i \cap A_j = \emptyset$.

Lemma 3.79 (Cardinality of a Finite Partition)**.** Let B be a finite set. Let $\{A_i\}_{i \in \mathbb{Z}_{1,k}}$ for $k \in \mathbb{Z}^+$ be a partition of B. Then

$$\sum_{i=1}^{k} |A_i| = |B|.$$

Proof. See Appendix C.15. □

Definition 3.80 (Euler's ϕ Function) — For $n \in \mathbb{Z}^+$, define Euler's ϕ function by

$$\phi(n) = |\{a \in \mathbb{Z}_{1,n} : \gcd(a, n) = 1\}|.$$

Definition 3.81 (Cyclic) — Let G be a finite set closed under multiplication. We say G is **cyclic** if there exists $g \in G$ such that

$$\mathrm{ord}(g) = |G|.$$

Lemma 3.82 (Full Order Implies Generation)**.** Let G be a finite set closed under multiplication, and let $g \in G$ satisfy

$$ord(g) = |G|.$$

Then

$$G = \{g^i : i \in [|G|]\}.$$

Proof. Let $d = |G| = ord(g)$, and define

$$H = \{g^i : i \in [d]\}.$$

Since $g \in G$ and G is closed under multiplication, every element of H lies in G (including g^0, since $g^0 = g^d = 1 \in G$), so

$$H \subseteq G.$$

We claim that the elements of H are distinct. Suppose

$$g^i = g^j$$

for some $i, j \in [d]$, WLOG with $i > j$. Then

$$g^{i-j} = 1.$$

But

$$0 < i - j < d,$$

contradicting $ord(g) = d$. Thus H has exactly d (distinct) elements, so

$$|H| = d = |G|.$$

Since $H \subseteq G$, Lemma 3.77 gives

$$H = G.$$

Therefore

$$G = \{g^i : i \in [|G|]\}.$$

□

3.7 Polynomials

Definition 3.83 (Polynomial) — For arbitrary ring R, define $R[x]$ to be the set of all sequences

$$(a_i)_{i\in\mathbb{Z}^{\geq 0}},$$

where $a_i \in R$ for every $i \in \mathbb{Z}^{\geq 0}$, such that there exists $N \in \mathbb{Z}^{\geq 0}$ such that for all $n \in \mathbb{Z}^{\geq 0}$, $n > N$, $a_n = 0$.

Let $f = (a_i)_{i\in\mathbb{Z}^{\geq 0}} \in R[x]$. We denote

$$f(x) = \sum_{i=0}^{n} a_i x^i,$$

where x is purely notational.

Definition 3.84 (Polynomial Addition) — Let $f, g \in R[x]$ be defined as

$$f = (a_i)_{i\in\mathbb{Z}^{\geq 0}},$$

$$g = (b_i)_{i\in\mathbb{Z}^{\geq 0}}.$$

Then define $f + g = (a_i + b_i)_{i\in\mathbb{Z}^{\geq 0}}$.

Lemma 3.85 (Well-Definedness of Polynomial Addition). For all $f, g \in R[x]$, we have

$$f + g \in R[x].$$

Thus polynomial addition is well-defined.

Proof. See Appendix D.1. □

Definition 3.86 (Polynomial Multiplication) — Let $f, g \in R[x]$ be defined as

$$f = (a_i)_{i\in\mathbb{Z}^{\geq 0}},$$

$$g = (b_i)_{i\in\mathbb{Z}^{\geq 0}}.$$

Then define $fg = (c_i)_{i\in\mathbb{Z}^{\geq 0}}$, where

$$c_n = \sum_{i=0}^{n} a_i b_{n-i},$$

for all $n \in \mathbb{Z}^{\geq 0}$.

Lemma 3.87 (Well-Definedness of Polynomial Multiplication). For all $f, g \in R[x]$, we have

$$fg \in R[x].$$

Thus polynomial multiplication is well-defined.

Proof. See Appendix D.2. □

Lemma 3.88 (Polynomial Ring)**.** $R[x]$ is a ring.

Proof. See Appendix D.3. $\square$

Definition 3.89 (Degree of Polynomials) — For a polynomial $g = (a_i)_{i \in \mathbb{Z}^{\geq 0}}$, let $\deg(g)$ denote the unique $n \in \mathbb{Z}^{\geq 0}$ such that $a_n \neq 0$ and $\forall j > n, a_j = 0$. The degree of the zero polynomial doesn't exist.

Lemma 3.90 (Existence and Uniqueness of Degree)**.** Every nonzero polynomial $g \in R[x]$ has a unique degree.

Proof. (Existence) Let $g = (a_i)_{i \in \mathbb{Z}^{\geq 0}}$ be nonzero. By the definition of a polynomial, there exists $N \in \mathbb{Z}^{\geq 0}$ such that, for every $k > N$,

$$a_k = 0.$$

Let

$$S = \{N \in \mathbb{Z}^{\geq 0} : \forall k > N,\ a_k = 0\}.$$

Then S is nonempty, so by WOP it has a least element M. Since $M \in S$,

$$a_k = 0$$

for every $k > M$.

It remains to show that $a_M \neq 0$. Assume for contradiction that $a_M = 0$.

If $M = 0$, then $a_0 = 0$, and since $a_k = 0$ for every $k > M = 0$, we have $a_k = 0$ for every $k \in \mathbb{Z}^{\geq 0}$. Thus $g = 0$, contradicting that g is nonzero.

Hence $M > 0$. By Lemma 3.18, $M \geq 1$, so $M - 1 \in \mathbb{Z}^{\geq 0}$. Since $a_M = 0$ and $a_k = 0$ for every $k > M$, we have

$$a_k = 0$$

for every $k > M - 1$. Therefore $M - 1 \in S$. But $M - 1 < M$, contradicting the minimality of M.

Thus $a_M \neq 0$. Since $a_k = 0$ for every $k > M$, we have

$$\deg(g) = M.$$

Therefore every nonzero polynomial has a degree. $\square$

Proof. (Uniqueness) Suppose $g = (a_i)_{i \in \mathbb{Z}^{\geq 0}}$ has two degrees $d_1, d_2 \in \mathbb{Z}^{\geq 0}$. Assume for contradiction that $d_1 \neq d_2$. WLOG let $d_1 < d_2$.

Since d_1 is a degree of g, we have

$$a_k = 0$$

for every $k > d_1$. In particular, because $d_2 > d_1$,

$$a_{d_2} = 0.$$

But since d_2 is also a degree of g,

$$a_{d_2} \neq 0,$$

a contradiction. Therefore $d_1 = d_2$ by contradiction, so the degree of g is unique. $\square$

Lemma 3.91 (Degree of Polynomial Product)**.** Let $p, q \in \mathbb{Z}_p[x]$ be nonzero polynomials. Then

$$\deg(pq) = \deg(p) + \deg(q).$$

Proof. Let

$$p = (p_i)_{i \in \mathbb{Z}^{\geq 0}} \qquad q = (q_i)_{i \in \mathbb{Z}^{\geq 0}}$$

, $\deg(p) = m$, $\deg(q) = n$. In the product $p \cdot q$,

1. The element $(p \cdot q)_k$, for $k > m + n$, is
$$\sum_{i=0}^{k} p_i q_{k-i}$$
If $i > m$, then $p_i = 0$. If $i \leq m$, then $k - i > (n + m) - i \geq n$, so $q_{k-i} = 0$. Thus the entire summation is equal to 0.

2. The element $(p \cdot q)_{m+n}$ is
$$\sum_{i=0}^{m+n} p_i q_{(m+n)-i}.$$
If $i > m$, then $p_i = 0$. If $i < m$, then $(m + n) - i > n$, so $q_{(m+n)-i} = 0$. Therefore this summation is equal to $p_m q_n$, both of which are nonzero by definition. As such, by Lemma 3.53, their product is also nonzero.

So the degree of $p \cdot q$ is $m + n$. ∎

Definition 3.92 (Polynomial Evaluation) — Let $f = (a_i)_{i \in \mathbb{Z}^{\geq 0}} \in R[x]$ and let $r \in R$.

If $\deg(f) = n$, we evaluate f at r as $f(r) = \sum_{i=0}^{n} a_i r^i$. If $f = 0$, then its evaluation is always 0.

Note that for each $k \geq n$,

$$\sum_{i=0}^{n} a_i r^i = \sum_{i=0}^{k} a_i r^i$$

as all future terms will be zero by the definition of degree.

Lemma 3.93 (Evaluation Homomorphism). Evaluation of polynomials $f, g \in R[x]$ at $r \in R$ respects addition and multiplication. That is,

$$f(r) + g(r) = (f+g)(r), \qquad f(r)g(r) = (f \cdot g)(r).$$

Proof. See Appendix D.4. $\square$

Definition 3.94 (Root) — For all $f \in R[x]$, we say $r \in R$ is a root of f if $f(r) = 0$.

Lemma 3.95 (Root Theorem). If $f(x) \in \mathbb{Z}_p[x]$ and $f(r) = 0$ for $r \in \mathbb{Z}_p$, then $f(x) = (x-r)q(x)$ for some $q(x) \in \mathbb{Z}_p[x]$.

Proof. If $f = 0$, then

$$f(x) = (x-r) \cdot 0,$$

so the result holds.

Now, let

$$S = \left\{ \deg(f) \in \mathbb{Z}^{\geq 0} : f \in \mathbb{Z}_p[x], f(r) = 0 \text{ and } \nexists q(x) \in \mathbb{Z}_p[x] \ s.t. \ f(x) = (x-r)q(x) \right\}.$$

Assume for the sake of contradiction that S is nonempty. Then, there exists a least element $m \in S$ by WOP. Choose a polynomial with $\deg(f) = m$, so that

$$f(x) = a_0 + a_1 x + \cdots + a_m x^m,$$

and where there exists no $q(x)$ such that $f(x) = (x-r)q(x)$.

We claim that $m \neq 0$. If $m = 0$, then

$$f(x) = a_0$$

with $a_0 \neq 0$. But then

$$f(r) = a_0 \neq 0,$$

contradicting $f(r) = 0$. Hence $m > 0$, and so by Lemma 3.18, $m \geq 1$.

Define

$$p(x) = f(x) - a_m x^{m-1}(x-r).$$

Then

$$\begin{aligned} p(x) &= f(x) - \left(a_m x^m - a_m r x^{m-1}\right) \\ &= f(x) - a_m x^m + a_m r x^{m-1}. \end{aligned}$$

Thus the coefficient of x^m in $p(x)$ is 0. Therefore either $p = 0$ or $\deg(p) < m$.

Also, note that by Lemma D.4,

$$\begin{aligned} p(r) &= f(r) - a_m r^{m-1}(r - r) \\ &= 0. \end{aligned}$$

We claim that $p \neq 0$. If $p = 0$, then

$$f(x) = a_m x^{m-1}(x - r) = (x - r)a_m x^{m-1},$$

contradicting our choice of f.

Hence $p \neq 0$, so $\deg(p) < m$. By the minimality of m in S, $\deg(p) \notin S$, so there exists $q_1(x) \in \mathbb{Z}_p[x]$ such that

$$p(x) = (x - r)q_1(x).$$

Thus

$$\begin{aligned} f(x) &= (x - r)q_1(x) + a_m x^{m-1}(x - r), \\ &= (x - r)\left(q_1(x) + a_m x^{m-1}\right). \end{aligned}$$

This contradicts our choice of f, so by contradiction, S is empty, and the result holds for all $f \in \mathbb{Z}_p[x]$. $\square$

4 Main Results

The dependencies among the following results and Section 3.6 are summarized in Figure 1. Node colors distinguish the "level" of each result (e.g. foundational lemmas, intermediate results, major claims, and the final theorem), and some boxes represent the combination of many lemmas. Dashed arrows indicate implicit dependencies, and thicker arrows indicate the claims part of the final theorem (4.9).

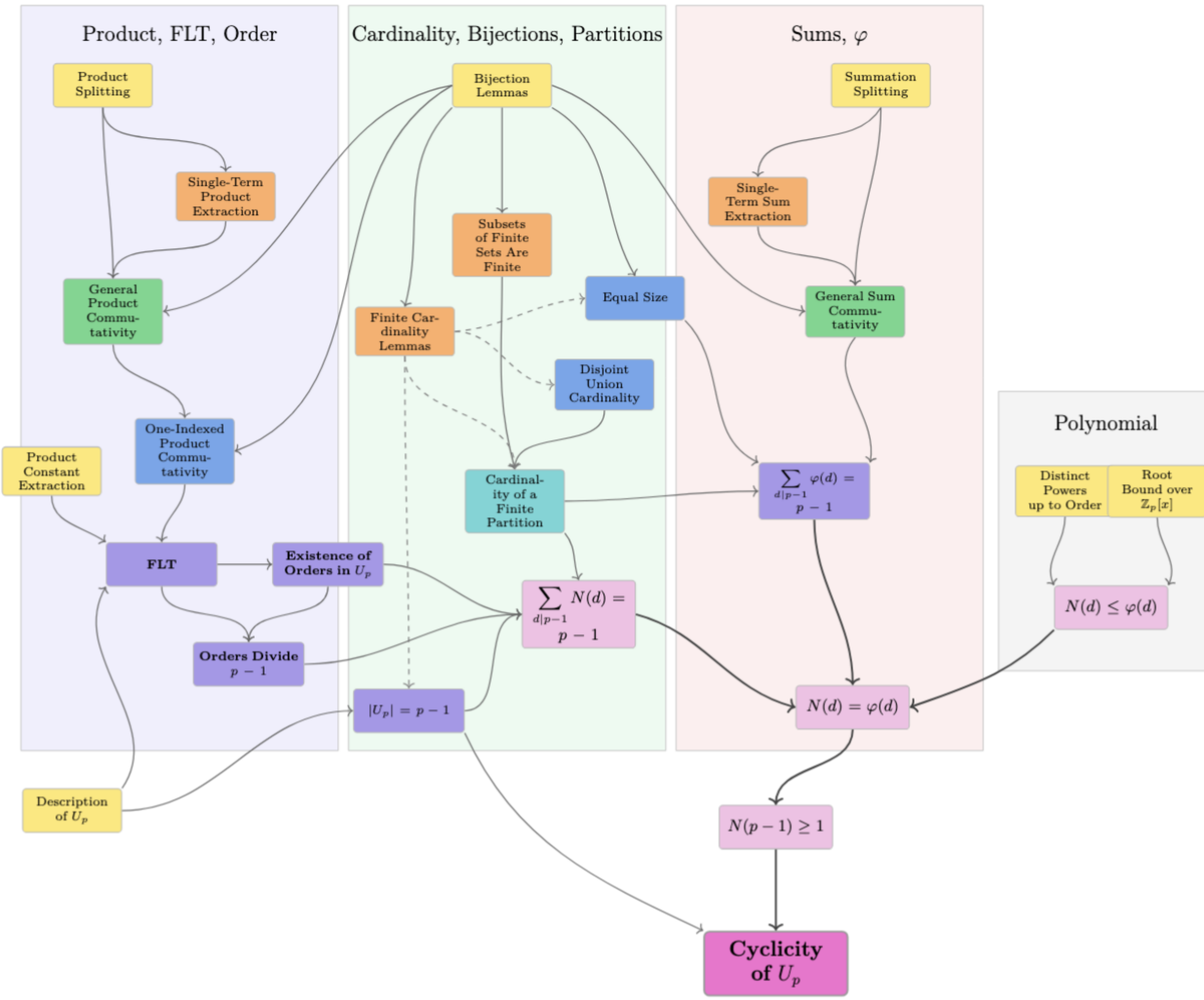

Figure 1: Dependency graph for 3.6 and Main Results leading up to the cyclicity of U_p.

Lemma 4.1 (Description of U_p)**.** For prime p,

$$U_p = \{[1], [2], \ldots, [p-1]\}.$$

Proof. Let

$$I_p = \{[1], [2], \ldots, [p-1]\}.$$

We show that $I_p \subseteq U_p$ and $U_p \subseteq I_p$.

Claim 4.1.1 — Let $[a] \in I_p$, where $1 \leq a < p$. Then $\gcd(a, p) = 1$.

Proof. Let $d = \gcd(a, p)$. Then $d \mid a$ and $d \mid p$. Since $d \in \mathbb{Z}^+$, we know $d > 0$, so by Lemma 3.18, $d \geq 1$.

Suppose for the sake of contradiction that $d > 1$. By Lemma 3.18, $d \geq 2$. Then since $d \mid p$, there exists $k \in \mathbb{Z}$ such that $dk = p$. Since $d > 0$ and $p > 0$, Lemma 3.21 gives $k > 0$. Hence $k \geq 1$.

If $k = 1$, then $d = p$. Since $d \mid a$, this gives $p \mid a$, so by Lemma 3.27, $p \leq a$, contradicting $a < p$.

Thus $k \neq 1$. Since $k \geq 1$, we have $k > 1$, and hence $k \geq 2$. Therefore $p = dk$ with $d, k \geq 2$, contradicting Lemma 3.30. Thus by contradiction we have $d = 1$, and so $\gcd(a, p) = 1$. □

Then by Bezout's Identity (Lemma 3.39), there exist $x, y \in \mathbb{Z}$ such that $ax + py = 1$. Hence

$$p \mid 1 - ax,$$

so

$$ax \equiv 1 \pmod{p}.$$

Therefore

$$[a][x] = [ax] = [1].$$

Thus $[a]$ has a multiplicative inverse in $\mathbb{Z}_p$, so $[a] \in U_p$. Therefore,

$$I_p \subseteq U_p.$$

For the other direction, let $u \in U_p$. Since $u \in \mathbb{Z}_p$, there exists $a \in \mathbb{Z}$ such that $u = [a]$. By the Division Algorithm (Lemma 3.34), there exist $q, r \in \mathbb{Z}$ such that

$$a = pq + r, \qquad 0 \leq r < p.$$

Thus $a - r = pq$, so $p \mid a - r$. Hence

$$[a] = [r],$$

and therefore

$$u = [r].$$

We claim that $r \neq 0$. Suppose for contradiction that $r = 0$. Then $u = [0]$. Since $u \in U_p$, there exists $v \in \mathbb{Z}_p$ such that

$$[0]v = [1].$$

But by Lemma 3.4,

$$[0]v = [0],$$

so $[0] = [1]$. By Lemma 3.42, this implies

$$0 \equiv 1 \pmod{p},$$

so $p \mid 1$, and by Lemma 3.27 we obtain $p \leq 1$, impossible for primes by definition. Thus $r \neq 0$.

Since $0 \leq r < p$ and $r \neq 0$, we have by Lemma 3.18,

$$1 \leq r < p.$$

Therefore $[r] \in I_p$, so $u \in I_p$. Hence
$$U_p \subseteq I_p.$$

Finally,
$$U_p = I_p = \{[1], [2], \ldots, [p-1]\}.$$

□

Lemma 4.2 (Cardinality of U_p)**.** For prime p,
$$|U_p| = p - 1.$$

Proof. By Lemma 4.1, $U_p = \{[1], [2], \ldots, [p-1]\}$. Define $f : [p-1] \to U_p$ by
$$f(i) = [i+1].$$

We show that f is a bijection.

Injective: Suppose $i, j \in [p-1]$ satisfy $f(i) = f(j)$. Then
$$[i+1] = [j+1].$$

By Lemma 3.42,
$$i + 1 \equiv j + 1 \pmod{p},$$

so
$$p \mid i - j.$$

Suppose for contradiction that $i \neq j$, and WLOG let $i > j$. Then $0 < i - j$. Since $i, j \in [p-1]$, we know $i < p$ and $-j \leq 0$. Hence $0 < i - j < p$. But $p \mid i - j$, so by Lemma 3.27,
$$p \leq i - j,$$

a contradiction. Hence $i = j$ by contradiction, so f is injective.

Surjective: Let $[a] \in U_p$. By Lemma 4.1, there exists $a \in \mathbb{Z}$ with $1 \leq a \leq p-1$ such that the given element is $[a]$. Then $a - 1 \in [p-1]$ and
$$f(a-1) = [a].$$

Therefore $f : [p-1] \to U_p$ is bijective, so by the definition of cardinality,
$$|U_p| = p - 1.$$

□

Lemma 4.3 (Root Bound for Polynomials over $\mathbb{Z}_p$)**.** Every nonzero $f(x) \in \mathbb{Z}_p[x]$ with degree d has at most d distinct roots.

Proof. Define the set

$$S = \left\{k \in \mathbb{Z}^{\geq 0} : \exists \text{ nonzero } f \in \mathbb{Z}_p[x], \text{ s.t.} \deg(f) = k \text{ and } f \text{ has more than } k \text{ distinct roots}\right\}.$$

As $S \subseteq \mathbb{Z}^{\geq 0}$, assume for contradiction that S is non-empty. By WOP, S has least element k. Choose the nonzero polynomial $f \in \mathbb{Z}_p[x]$ where f has more than $\deg(f) = k$ roots.

As each nonzero polynomial of degree 0 is a constant and thus has exactly 0 roots, f cannot be a polynomial of degree 0. Thus $k > 0$ so by Lemma 3.18, $k \geq 1$, so f must have at least one root. Let this root be r.

By the Root Theorem (Lemma 3.95, there exists

$$q \in \mathbb{Z}_p[x]$$

such that

$$f = (x - r)q.$$

Since $f \neq 0$, we have $q \neq 0$. Then, by Lemma 3.91,

$$\deg f = \deg{(x - r)} + \deg q,$$

so

$$\deg q = k - 1.$$

Since $k \geq 1$, $k - 1 \geq 0$. Also, $k - 1 < k$, so by the minimality of k in S, q has at most $k - 1$ distinct roots.

Now let c be an arbitrary root of f. Then

$$0 = f(c) = (c - r)q(c).$$

By Lemma 3.53, either $c - r = 0$ or $q(c) = 0$. Thus $c = r$, or c is a root of q. Therefore, the roots of f consist of at most the $k - 1$ roots of q, along with the single element r. Hence f has at most

$$(k - 1) + 1 = k$$

distinct roots, contradicting that $k \in S$. Therefore S is empty, and the lemma holds for all nonzero polynomials $f \in \mathbb{Z}_p[x]$. $\square$

Lemma 4.4 (Divisor Sum Formula for ϕ)**.** For each $n \in \mathbb{Z}^+$:

$$\sum_{d|n} \phi(d) = n$$

Proof. Let $D_n = \{d \in \mathbb{Z}^+ : d \mid n\}$. For each $d \in D_n$, let

$$X_d = \{a \in \mathbb{Z}^+ \mid 1 \leq a \leq n, \gcd(a, n) = d\}.$$

Claim 4.4.1 — $\{X_d\}_{d \in D_n}$ is a partition of $\mathbb{Z}_{1,n}$.

Proof. (i) We show that

$$\bigcup_{d \in D_n} X_d = \mathbb{Z}_{1,n}.$$

Let $a \in \mathbb{Z}_{1,n}$ be arbitrary. Thus we must show a is in some X_d. Since $a \geq 1$, $\gcd(a, n)$ exists. Let $d = \gcd(a, n)$, so then $d \mid n$, so $d \in D_n$. Hence, $a \in X_d$ by definition.

(ii) We show that all X_d are pairwise disjoint. Let $d, e \in \mathbb{Z}^+$ satisfy $d \mid n, e \mid n, d \neq e$. We claim that $X_d \cap X_e = \emptyset$. Assume for contradiction there exists $a \in X_d \cap X_e$. Then $\gcd(a, n) = d, \gcd(a, n) = e$. Thus since gcd is unique, $d = e$, a contradiction. Hence all distinct X_d are pairwise disjoint.

Therefore, $\{X_d\}_{d \in D_n}$ satisfies all properties of a partition of $\mathbb{Z}_{1,n}$. □

Then, by Lemma 3.79,

$$\sum_{d|n} |X_d| = |\mathbb{Z}_{1,n}| = n.$$

Claim 4.4.2 — For each $d \in D_n$, $|X_d| = \phi(n/d)$.

Proof. We show there is a bijection between

$$X_d = \{a \in \mathbb{Z}^+ \mid 1 \leq a \leq n, \gcd(a, n) = d\},$$

and

$$Y_d = \{b \in \mathbb{Z}^+ \mid 1 \leq b \leq n/d, \gcd(b, n/d) = 1\}.$$

Define $f : X_d \to Y_d$ by $f(a) = a/d$.

Claim — This function is well-defined.

Proof. Let $a \in X_d$. Then $\gcd(a, n) = d$, so $d \mid a, d \mid n$. We claim that $\gcd(a/d, n/d) = 1$.

We know that

$$1 \mid a/d, 1 \mid n/d$$

because both are integers, so property (i) of the gcd is satisfied.

Suppose $c \in \mathbb{Z}$ divides $a/d, n/d$. Then $cd \mid a, cd \mid n$. Thus, $cd \mid d$, so there exists $z \in \mathbb{Z}$ such that $(cd)(z) = d$. Thus,

$$(dc)(z) = d$$

$$d(cz) = d$$

$$d(cz - 1) = 0,$$

and since $d \neq 0$, by Lemma 3.19, $cz - 1 = 0$, so $cz = 1$. Thus $c \mid 1$, so 1 satisfies property (ii) of the gcd. Finally, since $1 > 0$, we are done. □

Claim — This function is injective.

Proof. Suppose $a_1, a_2 \in X_d$ and $f(a_1) = f(a_2)$. Then
$$\frac{a_1}{d} = \frac{a_2}{d}.$$
Multiplying both sides by d, we obtain
$$a_1 = a_2.$$
Thus f is injective. □

Claim — This function is surjective.

Proof. Let $b \in Y_d$. We claim that $a = bd \in X_d$. Since $1 \leq b \leq n/d$, and $d > 0$, we have
$$1 \leq d \leq bd \leq n.$$
Moreover,
$$\gcd(b, n/d) = 1.$$
We claim that $\gcd(bd, n) = d$.

(i) First, we see that $d \mid d$, so $d \mid bd$. Also, $d \mid n$ by definition.

(ii) Next, let $c \in \mathbb{Z}$ divide bd, n. Since $\gcd(b, n/d) = 1$, by Bezout's Identity (Lemma 3.39), there exists $x, y \in \mathbb{Z}$ such that
$$bx + (n/d)y = 1.$$
Hence,
$$\begin{aligned} d(bx + (n/d)y) &= d \cdot 1 \\ d(bx) + d((n/d)y) &= d \cdot 1 \\ (db)x + (d(n/d))y &= d \\ (bd)x + ny &= d. \end{aligned}$$
Also, $c \mid bd$, $c \mid n$, so by Lemma 3.26, c divides any linear combination of bd, n. Thus $c \mid d$.

(iii) $d > 0$, as $d \in \mathbb{Z}^+$ by definition.

Thus d satisfies all three properties of the gcd, so $\gcd(bd, n) = d$.
Hence, $bd \in X_d$. Moreover, $f(bd) = (bd)/d = b$, so f is surjective. □

Thus, f is bijective. Thus by Lemma 3.74,
$$|X_d| = |Y_d|.$$
But by definition, $|Y_d| = \phi(n/d)$. Hence $|X_d| = \phi(n/d)$. □

Therefore,
$$\sum_{d \mid n} \phi\left(\frac{n}{d}\right) = n.$$

Claim 4.4.3 — The function $f : D_n \to D_n$ defined by $f(d) = \frac{n}{d}$, is a bijection.

Proof. (Well-defined) First, f is well defined as $d \mid n$, so there exists $e \in \mathbb{Z}$ such that $de = n$ (recall we defined $n/d = e$). Moreover, $e > 0$ by Lemma 3.21, so $e \in \mathbb{Z}^+$ by definition. Then since $e \mid n$ and $e \in \mathbb{Z}^+$, we know $e \in D_n$.

(Injective) Say for $d_1, d_2 \in D_n$, $f(d_1) = f(d_2)$. Then $n/d_1 = n/d_2$, and since $n \neq 0$,

$$\begin{aligned}
d_1(n/d_1) &= d_1(n/d_2) \\
n &= d_1(n/d_2) \\
nd_2 &= (d_1(n/d_2))d_2 \\
nd_2 &= d_1((n/d_2)d_2) \\
nd_2 &= d_1 n \\
nd_2 &= nd_1 \\
d_2 &= d_1.
\end{aligned}$$

Thus f is injective.

(Surjective) Now let $y \in D_n$, and consider n/y. By above reasoning, $n/y \in D_n$. Moreover, by Lemma 3.25, $n/(n/y) = y$, so $f(n/y) = y$. Thus f is surjective.

Hence, f is bijective. $\square$

Thus, by Lemma 3.64,

$$\begin{aligned}
\sum_{d \mid n} \phi(d) &= \sum_{d \mid n} \phi(f(d)) \\
&= \sum_{d \mid n} \phi(n/d) \\
&= n,
\end{aligned}$$

as desired. $\square$

Theorem 4.5 (Fermat's Little Theorem)**.** For a positive prime $p \in \mathbb{Z}^+$, each $a \in U_p$ satisfies $a^{p-1} = 1$.

Proof. Let a be an arbitrary element of U_p. Then the function $f : U_p \to U_p$ defined by $f(n) = an$ is a bijection, which we will verify here:

1. Injectivity: For $x, y \in U_p$, let $f(x) = f(y)$. Thus $ax = ay$. As $a \in U_p$, by definition there exists $a^{-1} \in U_p$ with $a^{-1}a = 1$. Thus,

$$\begin{aligned} a^{-1}(ax) &= a^{-1}(ay) \\ (a^{-1}a)x &= (a^{-1}a)y \\ 1 \cdot x &= 1 \cdot y \\ x &= y \end{aligned}$$

   Hence f is injective.

2. Surjectivity: Pick arbitrary $x \in U_p$. Now, by closure, $a^{-1}x \in U_p$. Also, observe that

$$\begin{aligned} f(a^{-1}x) &= a(a^{-1}x) \\ &= (aa^{-1})x \\ &= (a^{-1}a)x \\ &= 1 \cdot x \\ &= x \end{aligned}$$

   Hence f is surjective.

By Lemma 4.1, we know that $U_p = \{[1], [2], \ldots, [p-1]\}$, and as $f : U_p \to U_p$ is a bijection, Lemma 3.66 yields

$$\prod_{i=1}^{p-1} [i] = \prod_{i=1}^{p-1} f([i]) = \prod_{i=1}^{p-1} a[i].$$

Then by Lemma 3.67,

$$\prod_{i=1}^{p-1} a[i] = a^{p-1} \prod_{i=1}^{p-1} [i].$$

As U_p is closed under multiplication, $\prod_{i=1}^{p-1}[i]$ is a unit, so let it have inverse $k \in U_p$.

$$\begin{aligned} k \prod_{i=1}^{p-1} [i] &= k \left(a^{p-1} \prod_{i=1}^{p-1} [i] \right) \\ k \prod_{i=1}^{p-1} [i] &= a^{p-1} \left(k \prod_{i=1}^{p-1} [i] \right) \\ 1 &= a^{p-1} \cdot 1 \\ 1 &= a^{p-1}, \end{aligned}$$

as desired. $\square$

Lemma 4.6 (Existence of Orders in U_p)**.** For every $a \in U_p$, the order $ord(a)$ exists.

Proof. Let $a \in U_p$, and let
$$S = \{n \in \mathbb{Z}^+ : a^n = 1\}.$$
By Fermat's Little Theorem (Lemma 4.5), $a^{p-1} = 1$, so $p - 1 \in S$ and S is nonempty.

Thus by WOP, S has least element n. Then
$$a^n = 1,$$
and we know that for every $m \in \mathbb{Z}^+$, if $a^m = 1$, then $n \leq m$. Therefore, by the definition of order,
$$ord(a) = n.$$
Hence the order of a exists. □

Lemma 4.7 (Orders Divide $p - 1$)**.** For each prime number $p \in \mathbb{Z}^+$, for each element $a \in U_p$, $ord(a) \mid (p - 1)$.

Proof. By Lemma 4.6, $ord(a)$ exists.

By the Division Algorithm (Lemma 3.34), there exist $q, r \in \mathbb{Z}$ such that
$$p - 1 = q \cdot ord(a) + r, \qquad 0 \leq r < ord(a).$$
Specifically, we know that $q > 0$, because if $q \leq 0$, then as orders are positive,
$$p - 1 = q \cdot ord(a) + r \leq 0 \cdot ord(a) + r = r.$$
But $r < ord(a)$, so $p - 1 < ord(a)$. By Fermat's Little Theorem (Lemma 4.5),
$$a^{p-1} = 1,$$
so by definition $ord(a) \leq p - 1$, a contradiction.

Now assume for the sake of contradiction that $r \neq 0$. Then
$$0 < r < ord(a).$$
By Fermat's Little Theorem (Lemma 4.5),
$$a^{p-1} = 1,$$
and since $a^{ord(a)} = 1$,
$$\begin{aligned} 1 &= a^{p-1} \\ &= a^{q \cdot ord(a) + r} \\ &= \left(a^{ord(a)}\right)^q \cdot a^r \\ &= 1^q \cdot a^r \\ &= a^r. \end{aligned}$$
Hence, $a^r = 1$, but $0 < r < ord(a)$, contradicting the definition of order.

Therefore $r = 0$ by contradiction, so $p - 1 = q \cdot ord(a)$ and hence $ord(a) \mid p - 1$. □

Lemma 4.8 (Distinct Powers up to Order)**.** For an element $g \in U_p$ with $ord(g) = d$, the set

$$G = \{g^i : i \in [d]\}$$

has d distinct elements.

Proof. Assume for the sake of contradiction that there exist $0 \leq i, j < d$, $i \neq j$, such that

$$g^i = g^j.$$

Without loss of generality, let $i > j$. Because $g^j \in U_p$, there exists an inverse $(g^j)^{-1} \in U_p$.

$$\begin{aligned} g^i &= g^j \\ g^i(g^j)^{-1} &= (g^j)(g^j)^{-1} \\ (g^i)(g^j)^{-1} &= 1. \end{aligned}$$

Since $i > j$, we can write $i = j + (i - j)$ with $i - j > 0$. Thus,

$$\begin{aligned} (g^i)(g^j)^{-1} &= (g^j g^{i-j})(g^j)^{-1} \\ &= (g^j(g^j)^{-1})(g^{i-j}) \\ &= 1(g^{i-j}) \\ &= g^{i-j}. \end{aligned}$$

Hence, $g^{i-j} = 1$.

But since $0 \leq j < i < d$, $-j \leq 0$, so $i - j \leq i < d$. Moreover, $j < i$, so $-j > -i$. Thus $i - j > i - i = 0$. Hence

$$0 < i - j < d,$$

which contradicts that g has order d.

Hence, no powers g^i, g^j are equal for distinct $i, j \in [d]$. Therefore, the elements of G are all pairwise distinct, so G has d distinct elements. □

Theorem 4.9 (Cyclicity of U_p)**.** Let $p \in \mathbb{Z}$ be a prime. Then, U_p is cyclic.

Proof. For each $d \in \mathbb{Z}^+$ such that $d \mid (p-1)$, let $N(d)$ denote the number of elements in U_p of order d.

Claim 4.9.1 — For every $d \mid p-1$,

$$N(d) \leq \phi(d).$$

Proof. We have two cases.

- Case 1: $N(d) = 0$. Then by definition of ϕ, $0 \leq \phi(d)$.
- Case 2: $N(d) > 0$. Then there must exist $g \in U_p$ such that $ord(g) = d$.

  By Lemma 4.3, the polynomial $x^d - 1$ has at most d roots over $\mathbb{Z}_p$. Now, since $ord(g) = d$, $g^d - 1 = 0$, so g is a root of $x^d - 1$.

  Consider the set of powers of g:

  $$G = \{g^i : i \in [d]\}$$

  As g has order d, this set has exactly d elements by Lemma 4.8. For each $i \in [d]$, we have:

  $$(g^i)^d = g^{id} = g^{di} = (g^d)^i = 1^i = 1.$$

  All d elements of G are roots of $x^d - 1$. Thus $x^d - 1$ has at least d roots. As before, $x^d - 1$ has at most d roots, so it has exactly d roots: the d elements of G.

  Also, let

  $$L = \{x \in \mathbb{Z}_{1,d} : \gcd(x, d) = 1\}.$$

  Then $|L| = \phi(d)$ by definition.

  Since $g^d = g^0 = 1$, we may also write

  $$G = \{g^x : x \in \mathbb{Z}_{1,d}\}.$$

  Let $h \in U_p$ have order d. Then $h^d = 1$ so h is a root of $x^d - 1$. By the above argument, $h = g^x$ for some $x \in \mathbb{Z}_{1,d}$.

  Suppose for contradiction $\gcd(x, d) \neq 1$. Since $\gcd(x, d) \in \mathbb{Z}^+$, we know $\gcd(x, d) > 1$. Let

  $$k = \frac{d}{\gcd(x, d)}.$$

  Then $k \in \mathbb{Z}^+$ and $k < d$. Moreover,

  $$\begin{aligned} (g^x)^k &= g^{xk} \\ &= g^{xd/\gcd(x,d)} \\ &= g^{d(x/\gcd(x,d))} \\ &= 1^{x/\gcd(x,d)} \\ &= 1. \end{aligned}$$

  Since $ord(g^x) = d$, we know $d \leq k$ by definition. But $k < d$, a contradiction. Thus by contradiction, $\gcd(x, d) = 1$, so $x \in L$.

  Therefore, every element of order d is of the form g^x for some $x \in L$. Since $|L| = \phi(d)$, there are at most $\phi(d)$ elements of order d. Thus

  $$N(d) \leq \phi(d).$$

□

Claim 4.9.2 — For every $d \mid p-1$, $N(d) = \phi(d)$.

Proof. For each $d \mid p-1, d \in \mathbb{Z}^+$, define $A_d = \{u \in U_p : ord(u) = d\}$. Then $|A_d| = N(d)$ by definition.

Claim — $\{A_d\}_{d \mid p-1}$ is a partition of U_p.

Proof. First we show that every element of U_p is contained in some A_d. Since every unit $u \in U_p$ has an order, let $d = ord(u)$. By Lemma 4.7, $d \mid p-1$, and $u \in A_d$. Thus,

$$\bigcup_{d \mid p-1} A_d = U_p.$$

Next we show that no two distinct A_d contain any shared elements. Suppose $d_1, d_2 \in \mathbb{Z}^+, d_1 \neq d_2$ satisfy $d_1, d_2 \mid p-1$. Assume for contradiction there exists

$$u \in A_{d_1} \cap A_{d_2}.$$

Then

$$ord(u) = d_1, ord(u) = d_2.$$

But since order is unique, $d_1 = d_2$, a contradiction. Thus $A_{d_1} \cap A_{d_2}$ is empty, so $\{A_d\}_{d \mid p-1}$ is a partition of U_p. □

Thus by Lemma 3.79 and Lemma 4.2,

$$\sum_{d \mid p-1} |A_d| = \sum_{d \mid p-1} N(d) = |U_p| = p-1.$$

Also, by Lemma 4.4,

$$\sum_{d \mid p-1} \phi(d) = p-1.$$

Hence $\sum_{d \mid p-1} N(d) = \sum_{d \mid p-1} \phi(d)$. But by Claim 4.9.1,

$$N(d) \leq \phi(d)$$

for every $d \mid p-1$. Suppose for contradiction that for some positive divisor d, we had $N(d) < \phi(d)$. Then,

$$\sum_{d \mid p-1} N(d) < \sum_{d \mid p-1} \phi(d),$$

a contradiction. Hence by contradiction, for every $d \mid p-1$, we have

$$N(d) = \phi(d).$$

□

Claim 4.9.3 — There exists an element of U_p of order $p-1$.

Proof. Since $p-1 \mid p-1$, by Claim 4 we have

$$N(p-1) = \phi(p-1).$$

Moreover, $\gcd(1, p-1) = 1$, therefore

$$\phi(p-1) \geq 1,$$

so

$$N(p-1) \geq 1.$$

Then by definition of $N(d)$, there must be some element with order $p-1$. □

By Lemma 4.2,

$$|U_p| = p-1.$$

By the preceding claim, there exists $g \in U_p$ such that

$$ord(g) = p-1 = |U_p|.$$

Therefore, by Definition 3.81, $\boxed{U_p \text{ is cyclic.}}$ □

A Basic Ring and Order Proofs

Lemma A.1 (Additive Inverse Identities). For all $a, b \in R$,

(i) $(-a) + a = 0$.

(ii) $-(-a) = a$.

(iii) $-a = a \cdot (-1) = (-1) \cdot a$.

(iv) $(-1)(-1) = 1$.

(v) $-(a + b) = (-a) + (-b)$.

(vi) $b - a = (-1)(a - b) = (a - b)(-1)$.

(vii) $a - (a - b) = b$.

(viii) $-(ab) = a(-b) = (-a)b$.

(ix) $(-a)(-b) = ab$.

Proof. (i) We have $(-a) + a = a + (-a) = 0$.

(ii) By definition,

$$-a + (-(-a)) = 0.$$

But $-a + a = 0$, so $-(-a) = a$ by Lemma 3.1.

(iii) We have

$$\begin{aligned} a + a \cdot (-1) &= a \cdot 1 + a \cdot (-1) \\ &= a(1 + (-1)) \\ &= a \cdot 0 \\ &= 0. \end{aligned}$$

Hence $a \cdot (-1) = -a$ by Lemma 3.2. By commutativity, $a \cdot (-1) = (-1) \cdot a$.

(iv) We see that $(-1)(-1) = -(-1)$ by (iii), and $-(-1) = 1$ by (ii).

(v) We know that

$$\begin{aligned} -(a + b) &= (-1)(a + b) \\ &= (-1)a + (-1)b \\ &= (-a) + (-b). \end{aligned}$$

(vi) We see that

$$\begin{aligned} (-1)(a - b) &= (-1)(a + (-b)) \\ &= (-1)(a) + (-1)(-b) \\ &= (-a) + (-(-b)) \\ &= (-a) + b \\ &= b + (-a) \\ &= b - a. \end{aligned}$$

Also, $(-1)(a - b) = (a - b)(-1)$ by commutativity.

(vii) We know

$$\begin{aligned}
a-(a-b) &= a+(-(a+(-b))) \\
&= a+(-1)(a+(-b)) \\
&= a+((-1)(a)+(-1)(-b)) \\
&= (a+(-1)(a))+(-1)(-b) \\
&= (a+(-a))+(-(-b)) \\
&= 0+(-(-b)) \\
&= (-(-b)) \\
&= b.
\end{aligned}$$

(viii) We see that

$$\begin{aligned}
-(ab) &= (-1)(ab) \\
&= [(-1)a]b \\
&= (-a)b
\end{aligned}$$

and

$$\begin{aligned}
-(ab) &= (ab)(-1) \\
&= a[b(-1)] \\
&= a(-b),
\end{aligned}$$

as desired.

(ix) We have

$$\begin{aligned}
(-a)(-b) &= [(-1)a][(-1)b] \\
&= [a(-1)][(-1)b] \\
&= a[(-1)[(-1)b]] \\
&= a[[(-1)(-1)]b] \\
&= a[1\cdot b] \\
&= ab.
\end{aligned}$$

□

Lemma A.2 (Basic Properties of Order). For all $x, y, z \in T$, the following hold:

(i) Exactly one of $x<y$, $x=y$, or $x>y$ is true.

(ii) If $x \geq y$ and $y \geq x$, then $x=y$.

(iii) If $x<y$ and $y<z$, then $x<z$.

(iv) If $x \leq y$ and $y \leq z$, then $x \leq z$.

(v) If $x<y$, then $x+z<y+z$.

(vi) If $z>0$ and $x<y$, then $xz<yz$.

(vii) If $z>0$ and $x \leq y$, then $xz \leq yz$.

(viii) If $x<0$, then $-x>0$. Similarly, if $x>0$, then $-x<0$.

Proof. (i) Consider $x - y \in T$. By Trichotomy, exactly one of $x - y \in P, x - y = 0, -(x - y) \in P$ is true.

- Case 1: $x - y \in P$. Then by definition, since $x = (x - y) + y$,

$$y + (x - y) = x,$$

so we have $y < x$, thus

$$x > y.$$

Now assume $x > y$. Then $y < x$, so there exists $h \in P$ such that $y + h = x$. But since

$$y + (x - y) = x,$$

and subtraction is unique,

$$x - y = h.$$

Thus $x - y \in P$.

- Case 2: $x - y = 0$. Then $x = (x - y) + y$, so

$$x = 0 + y.$$

Thus $x = y + 0$, so $x = y$. Now assume $x = y$. Then

$$x + (-y) = y + (-y),$$

so

$$x + (-y) = 0.$$

Thus $x - y = 0$.

- Case 3: $-(x - y) \in P$. Then

$$x = (x - y) + y,$$

so

$$\begin{aligned} -(x - y) + x &= (-(x - y)) + [(x - y) + y] \\ &= [(-(x - y)) + (x - y)] + y \\ &= 0 + y \\ &= y. \end{aligned}$$

Thus, $x + (-(x - y)) = y$. Therefore, $x < y$. Now assume $x < y$. Thus there exists $h \in P$ such that $x + h = y$. Then $x + (-(-h)) = y$, so

$$[x + (-(-h))] + (-h) = y + (-h)$$

$$x + [(-(-h)) + (-h)] = y + (-h)$$

$$x + 0 = y + (-h)$$

$$x = y + (-h).$$

Since subtraction is unique, $-h = x - y$. Thus $-(-h) = h = -(x - y)$. Therefore, $-(x - y) \in P$.

Since exactly one of the cases must hold, and each corresponding condition is equivalent, we have shown that exactly one of $x > y, x = y$, or $x < y$ is true.

(ii) If $x \geq y$, then either $x > y$ or $x = y$. Now, if $x > y$, then neither $y > x$ nor $y = x$ is true, so $y \geq x$ is false, which gives a contradiction. But if $x = y$, then $y = x$, so $y \geq x$ is true. Thus $x = y$.

(iii) If $x < y$, then there exists $k \in P$ such that $x + k = y$. Also, since $y < z$ there exists $m \in P$ such that $y + m = z$. Thus
$$\begin{aligned}(x+k)+m &= z\\ x+(k+m) &= z,\end{aligned}$$
and since P is closed under addition, $x < z$.

(iv) If $x \leq y$, we have two cases.

- Case 1: $x < y$. Then if $y < z$, we have $x < z$ by (iii). If $y = z$, then $x < z$ by substitution, so $x \leq z$.
- Case 2: $x = y$. Then $x \leq z$ by substitution.

Thus $x \leq z$ in all cases.

(v) We know there exists $p \in P$ such that $x + p = y$. Then
$$\begin{aligned}(x+p)+z &= y+z\\ x+(p+z) &= y+z\\ x+(z+p) &= y+z\\ (x+z)+p &= y+z.\end{aligned}$$
Thus by definition, $x + z < y + z$.

(vi) Since $x < y$, there exists $k \in P$ such that $x + k = y$. Then $z(x + k) = zy$. Thus
$$zx + zk = zy.$$
Since $z > 0, z \in P$, so $zk \in P$ by closure. Thus $zx < zy$, so $xz < yz$ by commutativity.

(vii) First, if $x < y$ then by (vi), we have $xz < yz$, so certainly $xz \leq yz$. Now, if $x = y$, then $xz = yz$, so $xz \leq yz$ as well.

(viii) First let $x < 0$. By Lemma 3.13, $x \notin P$, and since $x \neq 0$ by (i), $-x \in P$ by Trichotomy. Hence $-x > 0$ by Lemma 3.13. Now if $x > 0$, then $0 < x$. Hence by (v),
$$\begin{aligned}0+(-x) &< x+(-x)\\ -x &< x+(-x)\\ -x &< 0,\end{aligned}$$
as desired.

□

Lemma A.3 (Discreteness of $\mathbb{Z}$). The following hold:

(i) There is no $a \in \mathbb{Z}$ such that $0 < a < 1$.

(ii) If $a, m \in \mathbb{Z}$ and $a > m$, then $a \geq m + 1$.

Proof. (i) Let $S = \{n \in \mathbb{Z} : 0 < n < 1\}$. Since every $n \in S$ satisfies $n > 0$, S is a subset of $\mathbb{Z}^{\geq 0}$. Assume for contradiction that S is nonempty. Then by WOP, let it have least element $a \in S$. Because $a > 0$, then $a \in P$. Thus $a \cdot a \in P$ by closure, so $a \cdot a = a^2 \in P$. Thus $a^2 > 0$.

Also, because $a < 1$ and $a > 0$, $a \cdot a < a$ by Lemma 3.16. Then since $a < 1$, by Lemma 3.16, we have $a \cdot a < 1$.

Since $0 < a < 1$ and $a \cdot a \in \mathbb{Z}$ by closure, $a \cdot a \in S$. But since $a \cdot a < a$, this contradicts the minimality of a. Therefore, S is empty, so there is no such $a \in \mathbb{Z}$ such that $0 < a < 1$.

(ii) Since $a > m$, we know $a + (-m) > m + (-m)$ by 3.16, so $a + (-m) > 0$. By closure, $a + (-m) \in \mathbb{Z}$, so $a + (-m) < 1$ is impossible as there is no integer between 0 and 1. Thus $a + (-m) \geq 1$. Hence

$$[a + (-m)] + m \geq 1 + m,$$

so

$$a + [(-m) + m] \geq 1 + m,$$
$$a + 0 \geq 1 + m$$
$$a \geq m + 1,$$

as desired.

□

B Congruence Classes and Modular Arithmetic

Lemma B.1 (Linear Combination Divisibility)**.** If $a \mid b$ and $a \mid c$, then for any $x, y \in \mathbb{Z}$, $a \mid (bx + cy)$.

Proof. $a \mid b$ implies there exists $p \in \mathbb{Z}$ such that $ap = b$. Thus,

$$\begin{aligned} x(ap) &= xb \\ x(ap) &= bx \\ a(xp) &= bx. \end{aligned}$$

Likewise, there exists $q \in \mathbb{Z}$ such that $aq = c$, so using the same steps,

$$a(yq) = cy.$$

Adding these equalities yields

$$a(xp) + a(yq) = bx + cy.$$

By distributivity,

$$a(xp + yq) = bx + cy.$$

And by closure of addition and multiplication, $(xp + qy) \in \mathbb{Z}$. Therefore, $a \mid (bx + cy)$. □

Lemma B.2 (Exponent Properties)**.** Let $a \in R$. Then for all $m, n \in \mathbb{Z}^+$,

(i) $a^{m+n} = a^m \cdot a^n$.

(ii) $a^{mn} = (a^m)^n$.

Proof. 1. Let $m \in \mathbb{Z}^+$ be fixed. Let $S = \{n \in \mathbb{Z}^+ : a^m \cdot a^n \neq a^{m+n}\}$. Assume for the sake of contradiction that S is nonempty. By WOP, there exists a least element $k \in S$. This means k is the smallest positive integer where $a^m \cdot a^k \neq a^{m+k}$.

Let us test $k = 1$. Then, $a^1 = a^0 a = 1 \cdot a = a$. $a^m \cdot a^1 = a^m \cdot a = a^{m+1}$ by definition of exponentiation, thus $1 \notin S$. Since $k \in \mathbb{Z}^+$, we have $k \geq 1$ by Lemma 3.18. Since $k \neq 1$, we have $k > 1$. Thus $k - 1 > 0$, so $k - 1 \in \mathbb{Z}^+$. Since k is least in S, we have $k - 1 \notin S$. Therefore

$$a^m \cdot a^{k-1} = a^{m+(k-1)}.$$

Hence,

$$\begin{aligned} a^m \cdot a^k &= a^m \cdot (a^{k-1} \cdot a^1) \\ &= (a^m \cdot a^{k-1}) \cdot a \\ &= a^{m+(k-1)} \cdot a \\ &= a^{(m+k)-1} \cdot a \\ &= a^{m+k}. \end{aligned}$$

This contradicts $k \in S$. Therefore S is empty, so

$$a^m a^n = a^{m+n}$$

for all $n \in \mathbb{Z}^+$.

2. Let m be a fixed positive integer, and let

$$S = \{n \in \mathbb{Z}^+ : (a^m)^n \neq a^{mn}\}.$$

Assume for the sake of contradiction that S is nonempty. By WOP, there exists a least element $k \in S$. This means k is the smallest integer for which $(a^m)^k \neq a^{mk}$.

Let us test $k = 1$. $(a^m)^1 = a^m$. Also, $a^{m \cdot 1} = a^m$. So $1 \notin S$. As above, this yields $k > 1$, so $k - 1 > 0$, and thus $k - 1 \in \mathbb{Z}^+$. Since k is least in S, we have $k - 1 \notin S$. Therefore

$$(a^m)^{k-1} = a^{m(k-1)}.$$

Hence, by (i),

$$\begin{aligned} (a^m)^k &= (a^m)^{k-1} \cdot a^m \\ &= a^{mk-m} \cdot a^m \\ &= a^{(mk-m)+m} \\ &= a^{mk+(-m+m)} \\ &= a^{mk+0} \\ &= a^{mk}. \end{aligned}$$

This contradicts $k \in S$. Therefore S is empty, so

$$(a^m)^n = a^{mn}$$

for all $n \in \mathbb{Z}^+$.

□

Lemma B.3 (Bezout's Identity)**.** For all $a, b \in \mathbb{Z}$ with $(a, b) \neq (0, 0)$, there exist $x, y \in \mathbb{Z}$ such that

$$ax + by = \gcd(a, b).$$

Proof. In the proof of Lemma 3.36, the greatest common divisor was constructed as the least positive element of

$$S = \{ax + by : x, y \in \mathbb{Z},\ ax + by > 0\}.$$

As we showed S is nonempty, there must exist $x_0, y_0 \in \mathbb{Z}$ such that

$$\gcd(a, b) = ax_0 + by_0.$$

□

Lemma B.4 (Congruence Is an Equivalence Relation)**.** Let $m \in \mathbb{Z}$ with $m \neq 0$. Congruence modulo m is an equivalence relation on $\mathbb{Z}$.

Proof. (i) Reflexive: For an arbitrary $a \in \mathbb{Z}$, by Lemma 3.7, $a - a = a + (-a) = 0$. Additionally, by Lemma 3.31, $m \mid 0$, so $m \mid a - a$, implying that $a \equiv a \pmod{m}$.

(ii) Symmetric: We want to show that if $a \equiv b \pmod{m}$ then $b \equiv a \pmod{m}$.

Since $a \equiv b \pmod{m}$, $m \mid a - b$ and by definition, $a - b = mk$ for some $k \in \mathbb{Z}$. By Lemma 3.8, $b - a = (a - b)(-1)$, so

$$\begin{aligned} b - a &= (mk)(-1) \\ &= m(k(-1)). \end{aligned}$$

Therefore, $b - a = m(-k)$, so by definition, $m \mid b - a$, so $b \equiv a \pmod{m}$.

(iii) Transitive: We want to show that if $a \equiv b \pmod{m}$ and $b \equiv c \pmod{m}$ then $a \equiv c \pmod{m}$.

Since $a \equiv b \pmod{m}$, we have, $m \mid a - b$ and by definition, $a - b = mp$ for some $p \in \mathbb{Z}$. Similarly, $b - c = mq$ for some $q \in \mathbb{Z}$.

Summing, we obtain

$$(a - b) + (b - c) = mp + mq.$$

On the LHS, we have

$$\begin{aligned} (a - b) + (b - c) &= (a + (-b)) + (b + (-c)) \\ &= a + [(-b) + (b + (-c))] \\ &= a + [((-b) + b) + (-c)] \\ &= a + [0 + (-c)] \\ &= a + (-c). \end{aligned}$$

On the RHS, we have, $mp + mq = m(p + q)$.

Thus, $a + (-c) = m(p + q)$. By closure, $p + q \in \mathbb{Z}$, so by definition, $m \mid a + (-c)$. Also, by Lemma 3.7, $m \mid a - c$ so $a \equiv c \pmod{m}$.

$\square$

Lemma B.5 (Well-Definedness of Equivalence Class Addition)**.** The definition

$$[a] + [b] = [a + b]$$

for $[a], [b] \in \mathbb{Z}_p$ is well-defined.

Proof. Suppose that

$$[a] = [a'], \qquad [b] = [b'].$$

Then

$$a \equiv a' \pmod{p}, \qquad b \equiv b' \pmod{p}.$$

Hence

$$p \mid a - a', p \mid b - b'.$$

Then by Lemma 3.26,

$$p \mid (a - a') + (b - b').$$

But

$$(a-a')+(b-b')=(a+(-a'))+(b+(-b'))=(a+b)+((-a')+(-b'))=(a+b)-(a'+b').$$

Thus

$$a+b\equiv a'+b' \pmod p,$$

so

$$[a+b]=[a'+b'].$$

Hence addition is well-defined. □

Lemma B.6 (Well-Definedness of Equivalence Class Multiplication)**.** The definition

$$[a][b]=[ab]$$

for $[a],[b]\in\mathbb{Z}_p$ is well-defined.

Proof. Suppose that

$$[a]=[a'], \qquad [b]=[b'].$$

Then

$$a\equiv a' \pmod p, \qquad b\equiv b' \pmod p.$$

Hence

$$p\mid a-a', p\mid b-b'.$$

Now,

$$\begin{aligned}
ab-a'b' &= (ab-a'b')+0\\
&= (ab-a'b')+\big(a'b-(a'b)\big)\\
&= \big(ab+(-(a'b'))\big)+\big(a'b+(-(a'b))\big)\\
&= \big(ab+(-(a'b))\big)+\big((-(a'b'))+a'b\big)\\
&= \big(ab+(-a')b\big)+\big(a'(-b')+a'b\big)\\
&= b(a+(-a'))+a'((-b')+b)\\
&= b(a-a')+a'(b-b').
\end{aligned}$$

But by Lemma 3.26,

$$p\mid b(a-a')+a'(b-b').$$

Thus by substitution,

$$p\mid ab-a'b',$$

so

$$ab\equiv a'b' \pmod p,$$

and hence

$$[ab]=[a'b'].$$

Thus multiplication is well-defined. □

Lemma B.7 ($\mathbb{Z}_p$ Is a Ring). $\mathbb{Z}_p$ is a ring.

Proof. We verify each of the ring axioms:

1. Commutativity $(+)$: $[a] + [b] = [a + b]$. By the commutativity of $\mathbb{Z}$, $a + b = b + a$. Thus, $[a] + [b] = [b] + [a]$.

2. Associativity $(+)$: We want to show that $[a] + ([b] + [c]) = ([a] + [b]) + [c]$. We know $[b] + [c] = [b + c]$ and $[a] + [b + c] = [a + (b + c)]$. By the associativity of $\mathbb{Z}$, $a + (b + c) = (a + b) + c$. Thus, $[a + (b + c)] = [(a + b) + c] = [a + b] + [c] = ([a] + [b]) + [c]$.

3. Commutativity $(\cdot)$: We want to show that $[a][b] = [b][a]$. $[a][b] = [ab]$ and by the commutativity of $\mathbb{Z}$, $ab = ba$. Thus $[ab] = [ba]$, $[a][b] = [b][a]$.

4. Associativity $(\cdot)$: We want to show that $[a]([b][c]) = ([a][b])[c]$. We know $[b][c] = [bc]$ and $[a][bc] = [a(bc)]$. By the associativity of $\mathbb{Z}$, $a(bc) = (ab)c$. Thus, $[a]([b][c]) = [a(bc)] = [(ab)c] = [ab][c] = ([a][b])[c]$.

5. Distributivity: We want to show that $[a]([b] + [c]) = [a][b] + [a][c]$. $[b] + [c] = [b + c]$ and $[a][b + c] = [a(b + c)]$. By the distributivity of $\mathbb{Z}$, $a(b + c) = ab + ac$. Thus, $[a]([b] + [c]) = [a(b + c)] = [ab + ac] = [ab] + [ac] = [a][b] + [a][c]$.

6. Zero: Since $0 \in \mathbb{Z}$, by the definition of $\mathbb{Z}_p$, $[0] \in \mathbb{Z}_p$. Hence, we want to show that for every $[a] \in \mathbb{Z}_p$, $[a] + [0] = [a]$. $[a] + [0] = [a + 0]$. By the zero axiom in $\mathbb{Z}$, $a + 0 = a$. Thus, $[a] + [0] = [a + 0] = [a]$.

7. Negatives: We want to show that for every $[a] \in \mathbb{Z}_p$, there exists $[x] \in \mathbb{Z}_p$ such that $[a] + [x] = 0$. Let $x = [-a]$. Since $-a \in \mathbb{Z}$, by the definition of $\mathbb{Z}_p$, $[-a] \in \mathbb{Z}_p$. $[a] + [-a] = [a + (-a)]$. By the additive inverse axiom in $\mathbb{Z}$, $a + (-a) = 0$. Thus, $[a] + [-a] = [a + (-a)] = [0]$.

8. One: Since $1 \in \mathbb{Z}$, by the definition of $\mathbb{Z}_p$, $[1] \in \mathbb{Z}_p$. Hence, we want to show that for every $[a] \in \mathbb{Z}_p$, $[a][1] = [a]$. $[a][1] = [a(1)]$. By the one axiom in $\mathbb{Z}$, $a(1) = a$. Hence, $[a][1] = [a(1)] = [a]$.

Therefore, $\mathbb{Z}_p$ is a ring. $\square$

C Finite Sets, Sums, and Products

Lemma C.1 (Summation Splitting). Let $(a_i)_{i \in \mathbb{Z}^{\geq 0}}$ be a sequence of elements of a ring R. For all $m, n \in \mathbb{Z}^{\geq 0}$ satisfying $0 \leq n < m$,

$$\sum_{i=0}^{m} a_i = \sum_{i=0}^{n} a_i + \sum_{i=n+1}^{m} a_i.$$

Proof. Let $n \in \mathbb{Z}^{\geq 0}$ be fixed. Let

$$S = \left\{ m \in \mathbb{Z}^{\geq 0} : m > n \text{ and } \sum_{i=0}^{m} a_i \neq \sum_{i=0}^{n} a_i + \sum_{i=n+1}^{m} a_i \right\}.$$

Assume for contradiction that S is nonempty, so by WOP it has a least element m.

We first have $m \neq n + 1$, since by the definition of summation,

$$\sum_{i=0}^{n+1} a_i = \sum_{i=0}^{n} a_i + a_{n+1} = \sum_{i=0}^{n} a_i + \sum_{i=n+1}^{n+1} a_i.$$

Therefore $m > n+1$ and so $m-1 > n$. Since m is the least element of S, we have $m-1 \notin S$, so the statement holds for $m-1$:

$$\sum_{i=0}^{m-1} a_i = \sum_{i=0}^{n} a_i + \sum_{i=n+1}^{m-1} a_i.$$

Again by definition,

$$\begin{aligned}
\sum_{i=0}^{m} a_i &= \left(\sum_{i=0}^{m-1} a_i\right) + a_m \\
&= \left(\sum_{i=0}^{n} a_i + \sum_{i=n+1}^{m-1} a_i\right) + a_m \\
&= \sum_{i=0}^{n} a_i + \left(\sum_{i=n+1}^{m-1} a_i + a_m\right) \\
&= \sum_{i=0}^{n} a_i + \sum_{i=n+1}^{m} a_i.
\end{aligned}$$

This contradicts $m \in S$. Therefore S is empty, so the lemma is true in general. □

Lemma C.2 (Single-Term Sum Extraction)**.** Let $(a_i)_{i\in\mathbb{Z}^{\geq 0}}$ be a sequence in a ring R. If $0 < m < n$, then

$$\sum_{i=0}^{n} a_i = a_m + \left(\sum_{i=0}^{m-1} a_i + \sum_{i=m+1}^{n} a_i\right).$$

Proof. By Lemma C.1,

$$\sum_{i=0}^{n} a_i = \sum_{i=0}^{m} a_i + \sum_{i=m+1}^{n} a_i.$$

By definition of summation on the first sum,

$$\sum_{i=0}^{m} a_i = \left(\sum_{i=0}^{m-1} a_i\right) + a_m$$

Therefore,

$$\begin{aligned}
\sum_{i=0}^{n} a_i &= \left(\left(\sum_{i=0}^{m-1} a_i\right) + a_m\right) + \sum_{i=m+1}^{n} a_i \\
&= a_m + \left(\sum_{i=0}^{m-1} a_i + \sum_{i=m+1}^{n} a_i\right),
\end{aligned}$$

by associativity and commutativity of addition. □

Lemma C.3 (Product Splitting). Let $(a_i)_{i\in\mathbb{Z}^{\geq 0}}$ be a sequence of elements of a ring R. For all $m,n\in\mathbb{Z}^{\geq 0}$ satisfying $0\leq n<m$,

$$\prod_{i=0}^{m} a_i = \left(\prod_{i=0}^{n} a_i\right)\left(\prod_{i=n+1}^{m} a_i\right).$$

Proof. Let $n\in\mathbb{Z}^{\geq 0}$ be fixed. Let

$$S=\left\{m\in\mathbb{Z}^{\geq 0} : m>n \text{ and } \prod_{i=0}^{m} a_i \neq \left(\prod_{i=0}^{n} a_i\right)\left(\prod_{i=n+1}^{m} a_i\right)\right\}.$$

Assume for contradiction that S is nonempty, so by WOP it has a least element m.

We first have $m\neq n+1$, since by the definition of product,

$$\prod_{i=0}^{n+1} a_i = \left(\prod_{i=0}^{n} a_i\right)a_{n+1} = \left(\prod_{i=0}^{n} a_i\right)\left(\prod_{i=n+1}^{n+1} a_i\right).$$

Therefore $m>n+1$ and so $m-1>n$. Since m is the least element of S, we have $m-1\notin S$, so the statement holds for $m-1$:

$$\prod_{i=0}^{m-1} a_i = \left(\prod_{i=0}^{n} a_i\right)\left(\prod_{i=n+1}^{m-1} a_i\right).$$

Again by definition,

$$\begin{aligned}\prod_{i=0}^{m} a_i &= \left(\prod_{i=0}^{m-1} a_i\right)a_m\\ &= \left[\left(\prod_{i=0}^{n} a_i\right)\left(\prod_{i=n+1}^{m-1} a_i\right)\right]a_m\\ &= \left(\prod_{i=0}^{n} a_i\right)\left[\left(\prod_{i=n+1}^{m-1} a_i\right)a_m\right]\\ &= \left(\prod_{i=0}^{n} a_i\right)\left(\prod_{i=n+1}^{m} a_i\right).\end{aligned}$$

This contradicts $m\in S$. Therefore S is empty, so the lemma is true in general. □

Lemma C.4 (Single-Term Product Extraction). Let $(a_i)_{i\in\mathbb{Z}^{\geq 0}}$ be a sequence in a ring R. For all $m,n\in\mathbb{Z}^{\geq 0}$ satisfying $0<m<n$,

$$\prod_{i=0}^{n} a_i = a_m\left(\prod_{i=0}^{m-1} a_i\right)\left(\prod_{i=m+1}^{n} a_i\right).$$

Proof. By Lemma C.3,

$$\prod_{i=0}^{n} a_i = \left(\prod_{i=0}^{m} a_i\right)\left(\prod_{i=m+1}^{n} a_i\right).$$

By definition of product notation on the first product,

$$\prod_{i=0}^{m} a_i = \left(\prod_{i=0}^{m-1} a_i\right)a_m$$

Therefore,

$$\begin{aligned}\prod_{i=0}^{n} a_i &= \left(\left(\prod_{i=0}^{m-1} a_i\right) a_m\right)\left(\prod_{i=m+1}^{n} a_i\right) \\ &= a_m \left(\prod_{i=0}^{m-1} a_i\right)\left(\prod_{i=m+1}^{n} a_i\right),\end{aligned}$$

by associativity and commutativity of multiplication. ☐

Lemma C.5 (Bijection Composition)**.** Let A, B, C be sets. If $f : A \to B$ and $g : B \to C$ are bijections, then $h = g \circ f : A \to C$, defined by $h(x) = g(f(x))$, is also a bijection.

Proof. (Injective) Let $x_1, x_2 \in A$ be arbitrary such that $h(x_1) = h(x_2)$. Then

$$g(f(x_1)) = g(f(x_2)),$$

so by injectivity of g,

$$f(x_1) = f(x_2).$$

Also, by injectivity of f,

$$x_1 = x_2,$$

as desired.

(Surjective) Let $c \in C$ be arbitrary. By surjectivity of g, there exists $b \in B$ such that $g(b) = c$. By surjectivity of f, there exists $a \in A$ such that $f(a) = b$. Then

$$h(a) = g(f(a)) = g(b) = c,$$

as desired.

Thus h is bijective. ☐

Lemma C.6 (General Commutativity of Sums)**.** Let $n \in \mathbb{Z}^{\geq 0}$. For a sequence $(a_i)_{i\in\mathbb{Z}^{\geq 0}}$, let $f : [n+1] \to [n+1]$ be a bijection. Then

$$\sum_{i=0}^{n} a_i = \sum_{i=0}^{n} a_{f(i)}.$$

Proof. Let

$$S = \left\{ n \in \mathbb{Z}^{\geq 0} : \exists (a_i)_{i\in\mathbb{Z}^{\geq 0}} \text{ and bijection } f : [n+1] \to [n+1], \text{ such that } \sum_{i=0}^{n} a_i \neq \sum_{i=0}^{n} a_{f(i)} \right\}.$$

Assume for contradiction that S is nonempty. By WOP, S has least element n. If $n = 0$, then $[n+1] = [1] = \{0\}$, so $f(0) = 0$. Therefore,

$$\sum_{i=0}^{0} a_{f(i)} = a_{f(0)} = a_0 = \sum_{i=0}^{0} a_i,$$

a contradiction. Thus $n \neq 0$, so $n > 0$. But $n < 1$ is impossible as there is no integer between 0 and 1, thus, $n \geq 1$. Hence $n - 1 \geq 0$.

Choose a sequence $(a_i)_{i\in\mathbb{Z}^{\geq 0}}$ and a bijection $f : [n+1] \to [n+1]$ for which the equality fails. Since f is surjective, there exists $k \in [n+1]$ such that

$$f(k) = n,$$

which is unique by injectivity.

Now we construct a new bijection from $[n] \to [n]$. By Lemma 3.63,

$$f' : [n+1] \setminus \{k\} \to [n]$$

defined by $f'(i) = f(i)$ for all $i \in [n+1] \setminus \{k\}$ is a bijection. Then we define

$$g : [n] \to [n+1] \setminus \{k\}$$

by

$$g(i) = \begin{cases} i, & i < k, \\ i+1, & i \geq k. \end{cases}$$

Claim C.6.1 — g is a bijection.

Proof. (Injective) Assume $a, b \in [n]$ such that $g(a) = g(b)$. If $a, b < k$, then $g(a) = a$ and $g(b) = b$, so $a = b$.

We claim that the case where $a < k, b \geq k$, or symmetrically, $a \geq k, b < k$, is a contradiction. Without loss of generality, assume $a < k, b \geq k$. Then $g(a) = a, g(b) = b + 1$, so $a = b + 1$. Since $1 \in P$, $b < a$. Thus $b < k$ by transitivity, contradicting $b \geq k$. Thus we have a contradiction.

Finally, if $a, b \geq k$, then $g(a) = a + 1, g(b) = b + 1$. Thus $a + 1 = b + 1$, so $a = b$. Thus g is injective.

(Surjective) Let $y \in [n+1] \setminus \{k\}$ be arbitrary.

If $y < k$, then $y < n$, so $y \in [n]$. Also, $g(y) = y$. However, if $y > k$, then $y \leq n$, so $y - 1 \leq n - 1$. Thus $y - 1 \in [n]$. Also, $y - 1 > k - 1$, so $y - 1 \geq k$ by Lemma 3.18. Thus, $g(y-1) = (y-1) + 1 = y$. ☐

Now define $h = f' \circ g$. By Lemma 3.62, $h : [n] \to [n]$ is a bijection. Since $0 \leq n - 1 < n$, and n is minimal in S, then $n - 1 \notin S$. Then, we have

$$\sum_{i=0}^{n-1} a_i = \sum_{i=0}^{n-1} a_{h(i)}.$$

Case 1: If $0 < k < n$, then by definition of g and h,

$$\sum_{i=0}^{n-1} a_{h(i)} = \sum_{i=0}^{k-1} a_{f(i)} + \sum_{i=k}^{n-1} a_{f(i+1)} = \sum_{i=0}^{k-1} a_{f(i)} + \sum_{j=k+1}^{n} a_{f(j)}.$$

By Lemma C.2,

$$\sum_{j=0}^{n} a_{f(j)} = a_{f(k)} + \left(\sum_{j=0}^{k-1} a_{f(j)} + \sum_{j=k+1}^{n} a_{f(j)} \right).$$

Therefore,

$$\sum_{j=0}^{n} a_{f(j)} = a_{f(k)} + \sum_{i=0}^{n-1} a_{h(i)}.$$

Case 2: Suppose $k = 0$. Then for every $i \in [n]$, we have $g(i) = i + 1$, so

$$h(i) = f'(g(i)) = f'(i+1) = f(i+1).$$

Therefore,

$$\sum_{i=0}^{n-1} a_{h(i)} = \sum_{i=0}^{n-1} a_{f(i+1)} = \sum_{j=1}^{n} a_{f(j)}.$$

By Lemma C.1,

$$\sum_{j=0}^{n} a_{f(j)} = a_{f(0)} + \sum_{j=1}^{n} a_{f(j)}.$$

Since $k = 0$, then,

$$\sum_{j=0}^{n} a_{f(j)} = a_{f(k)} + \sum_{i=0}^{n-1} a_{h(i)}.$$

Case 3: Suppose $k = n$. Then for every $i \in [n]$, we have $i < k$, so $g(i) = i$. Hence

$$h(i) = f'(g(i)) = f'(i) = f(i).$$

Therefore,

$$\sum_{i=0}^{n-1} a_{h(i)} = \sum_{i=0}^{n-1} a_{f(i)}.$$

By the definition of summation,

$$\sum_{j=0}^{n} a_{f(j)} = \left(\sum_{j=0}^{n-1} a_{f(j)} \right) + a_{f(n)}.$$

Since $k = n$,

$$\sum_{j=0}^{n} a_{f(j)} = \left(\sum_{i=0}^{n-1} a_{h(i)} \right) + a_{f(k)} = a_{f(k)} + \sum_{i=0}^{n-1} a_{h(i)}.$$

Hence, in all cases,

$$\sum_{j=0}^{n} a_{f(j)} = a_{f(k)} + \sum_{i=0}^{n-1} a_{h(i)}.$$

Since $f(k) = n$, we have $a_{f(k)} = a_n$. Also, since $n - 1 \notin S$,

$$\sum_{i=0}^{n-1} a_i = \sum_{i=0}^{n-1} a_{h(i)}.$$

Hence,

$$\begin{aligned}
\sum_{j=0}^{n} a_{f(j)} &= a_n + \sum_{i=0}^{n-1} a_{h(i)} \\
&= a_n + \sum_{i=0}^{n-1} a_i \\
&= \sum_{i=0}^{n} a_i,
\end{aligned}$$

which contradicts our choice of (a_i) and f. Hence S is empty, and therefore the lemma holds for all $n \in \mathbb{Z}^{\geq 0}$, $(a_i)_{i \in \mathbb{Z}^{\geq 0}}$, and bijections $f : [n+1] \to [n+1]$. $\square$

Lemma C.7 (General Product Commutativity)**.** Let $n \in \mathbb{Z}^+$. For a sequence $(a_i)_{i\in\mathbb{Z}^{\geq 0}}$, let $f : [n] \to [n]$ be a bijection. Then
$$\prod_{i=0}^{n-1} a_i = \prod_{i=0}^{n-1} a_{f(i)}.$$

Proof. Let
$$S = \left\{ n \in \mathbb{Z}^+ : \exists (a_i)_{i\in\mathbb{Z}^{\geq 0}} \text{ and bijection } f : [n] \to [n], \text{ such that } \prod_{i=0}^{n-1} a_i \neq \prod_{i=0}^{n-1} a_{f(i)} \right\}.$$

Assume for contradiction that S is nonempty. By WOP, S has least element n. If $n = 1$, then $[n] = [1] = \{0\}$, so $f(0) = 0$. Therefore,
$$\prod_{i=0}^{0} a_{f(i)} = a_{f(0)} = a_0 = \prod_{i=0}^{0} a_i,$$
a contradiction. Thus $n \neq 1$, so $n > 1$. Hence $n - 1 \in \mathbb{Z}^+$.

Choose a sequence $(a_i)_{i\in\mathbb{Z}^{\geq 0}}$ and a bijection $f : [n] \to [n]$ for which the equality fails. Since f is surjective, there exists $k \in [n]$ such that
$$f(k) = n - 1,$$
which is unique by injectivity.

Now we construct a new bijection from $[n-1] \to [n-1]$. By Lemma 3.63,
$$f' : [n] \setminus \{k\} \to [n-1]$$
defined by $f'(i) = f(i)$ for all $i \in [n] \setminus \{k\}$ is a bijection. Then we define
$$g : [n-1] \to [n] \setminus \{k\}$$
by
$$g(i) = \begin{cases} i, & i < k, \\ i+1, & i \geq k. \end{cases}$$
As in the proof of Lemma 3.64, g is a bijection.

Now define $h = f' \circ g$. By Lemma 3.62, $h : [n-1] \to [n-1]$ is a bijection. Since $n - 1 < n$, and n is minimal in S, then $n - 1 \notin S$. Then, we have
$$\prod_{i=0}^{n-2} a_i = \prod_{i=0}^{n-2} a_{h(i)}.$$

Case 1: If $0 < k < n-1$, then by definition of g and h,
$$\prod_{i=0}^{n-2} a_{h(i)} = \left(\prod_{i=0}^{k-1} a_{f(i)} \right) \left(\prod_{i=k}^{n-2} a_{f(i+1)} \right) = \left(\prod_{i=0}^{k-1} a_{f(i)} \right) \left(\prod_{j=k+1}^{n-1} a_{f(j)} \right).$$

By Lemma C.4,
$$\prod_{j=0}^{n-1} a_{f(j)} = a_{f(k)} \left(\prod_{j=0}^{k-1} a_{f(j)} \right) \left(\prod_{j=k+1}^{n-1} a_{f(j)} \right).$$

Therefore,
$$\prod_{j=0}^{n-1} a_{f(j)} = a_{f(k)} \prod_{i=0}^{n-2} a_{h(i)}.$$

Case 2: Suppose $k = 0$. Then for every $i \in [n-1]$, we have $g(i) = i + 1$, so

$$h(i) = f'(g(i)) = f'(i+1) = f(i+1).$$

Therefore,

$$\prod_{i=0}^{n-2} a_{h(i)} = \prod_{i=0}^{n-2} a_{f(i+1)} = \prod_{j=1}^{n-1} a_{f(j)}.$$

By Lemma C.3,

$$\prod_{j=0}^{n-1} a_{f(j)} = a_{f(0)} \prod_{j=1}^{n-1} a_{f(j)}.$$

Since $k = 0$, then,

$$\prod_{j=0}^{n-1} a_{f(j)} = a_{f(k)} \prod_{i=0}^{n-2} a_{h(i)}.$$

Case 3: Suppose $k = n - 1$. Then for every $i \in [n-1]$, we have $i < k$, so $g(i) = i$. Hence

$$h(i) = f'(g(i)) = f'(i) = f(i).$$

Therefore,

$$\prod_{i=0}^{n-2} a_{h(i)} = \prod_{i=0}^{n-2} a_{f(i)}.$$

By the definition of product,

$$\prod_{j=0}^{n-1} a_{f(j)} = \left(\prod_{j=0}^{n-2} a_{f(j)}\right) a_{f(n-1)}.$$

Since $k = n - 1$,

$$\prod_{j=0}^{n-1} a_{f(j)} = \left(\prod_{i=0}^{n-2} a_{h(i)}\right) a_{f(k)} = a_{f(k)} \prod_{i=0}^{n-2} a_{h(i)}.$$

Hence, in all cases,

$$\prod_{j=0}^{n-1} a_{f(j)} = a_{f(k)} \prod_{i=0}^{n-2} a_{h(i)}.$$

Since $f(k) = n - 1$, we have $a_{f(k)} = a_{n-1}$. Also, since $n - 1 \notin S$,

$$\prod_{i=0}^{n-2} a_i = \prod_{i=0}^{n-2} a_{h(i)}.$$

Hence,

$$\begin{aligned}
\prod_{j=0}^{n-1} a_{f(j)} &= a_{n-1} \prod_{i=0}^{n-2} a_{h(i)} \\
&= a_{n-1} \prod_{i=0}^{n-2} a_i \\
&= \prod_{i=0}^{n-1} a_i,
\end{aligned}$$

which contradicts our choice of (a_i) and f. Hence S is empty, and therefore the lemma holds for all $n \in \mathbb{Z}^+$, sequences $(a_i)_{i \in \mathbb{Z}^{\geq 0}}$, and bijections $f : [n] \to [n]$. $\square$

Lemma C.8 (One-Indexed Product Commutativity)**.** Let $n \in \mathbb{Z}^+$. For a sequence $(a_i)_{i \in \mathbb{Z}^{\geq 0}}$, let $f : \mathbb{Z}_{1,n} \to \mathbb{Z}_{1,n}$ be a bijection. Then

$$\prod_{i=1}^{n} a_i = \prod_{i=1}^{n} a_{f(i)}.$$

Proof. Define

$$s : [n] \to \mathbb{Z}_{1,n} \qquad s(i) = i + 1, \qquad t : \mathbb{Z}_{1,n} \to [n] \qquad t(j) = j - 1.$$

We can verify that both s and t are bijections. Since f is also a bijection, by Lemma 3.62,

$$g = t \circ f \circ s : [n] \to [n]$$

is also a bijection. In particular,

$$g(i) = f(i + 1) - 1.$$

Let $b_i = a_{i+1}$ for each $i \in \mathbb{Z}^{\geq 0}$. By Lemma 3.65,

$$\prod_{i=0}^{n-1} b_i = \prod_{i=0}^{n-1} b_{g(i)}.$$

Then, since $b_{g(i)} = a_{g(i)+1} = a_{(f(i+1)-1)+1} = a_{f(i+1)}$, reindexing by $j = i + 1$ gives

$$\prod_{j=1}^{n} a_j = \prod_{j=1}^{n} a_{f(j)},$$

as desired. $\square$

Lemma C.9 (Product Constant Extraction)**.** Let $a \in R$ and $n \in \mathbb{Z}^+$. Then

$$\prod_{i=1}^{n} ai = a^n \prod_{i=1}^{n} i.$$

Proof. Let

$$S = \left\{ n \in \mathbb{Z}^+ : \prod_{i=1}^{n} ai \neq a^n \prod_{i=1}^{n} i \right\}.$$

Assume for contradiction that S is nonempty, so by WOP it has a least element n.

We first have $n \neq 1$, since by the definition of product,

$$\prod_{i=1}^{1} ai = a = a^1 \prod_{i=1}^{1} i.$$

Therefore $n > 1$, so $n - 1 \in \mathbb{Z}^+$. Since n is the least element of S, we have $n - 1 \notin S$, so the statement holds for $n - 1$:

$$\prod_{i=1}^{n-1} ai = a^{n-1} \prod_{i=1}^{n-1} i.$$

Again by definition,

$$\begin{aligned}\prod_{i=1}^{n} ai &= \left(\prod_{i=1}^{n-1} ai\right)(an)\\ &= \left(a^{n-1}\prod_{i=1}^{n-1} i\right)(an)\\ &= a^n\left(\prod_{i=1}^{n-1} i\right)n\\ &= a^n\prod_{i=1}^{n} i.\end{aligned}$$

This contradicts $n \in S$. Therefore S is empty, so the lemma is true in general. $\square$

Lemma C.10 (Linearity of Finite Sums). Let $(x_i)_{i\in\mathbb{Z}^{\geq 0}}$ and $(y_i)_{i\in\mathbb{Z}^{\geq 0}}$ be sequences in a ring R. Then, for all $n \in \mathbb{Z}^{\geq 0}$,

$$\sum_{i=0}^{n}(x_i + y_i) = \sum_{i=0}^{n} x_i + \sum_{i=0}^{n} y_i.$$

Proof. Let

$$S = \left\{n \in \mathbb{Z}^{\geq 0} : \sum_{i=0}^{n}(x_i + y_i) \neq \sum_{i=0}^{n} x_i + \sum_{i=0}^{n} y_i\right\}.$$

Assume for contradiction that S is nonempty, so by WOP it has least element m.

First, we claim $m \neq 0$. Indeed, if $m = 0$, then

$$\sum_{i=0}^{0}(x_i + y_i) = x_0 + y_0 = \sum_{i=0}^{0} x_i + \sum_{i=0}^{0} y_i,$$

contradicting $m \in S$.

Thus, $m > 0$, so by Lemma 3.18, $m \geq 1$. Hence $m - 1 \in \mathbb{Z}^{\geq 0}$, and $m - 1 < m$. By minimality of m, then, the statement must hold true for $m - 1$:

$$\sum_{i=0}^{m-1}(x_i + y_i) = \sum_{i=0}^{m-1} x_i + \sum_{i=0}^{m-1} y_i.$$

Then by definition,

$$\begin{aligned}\sum_{i=0}^{m}(x_i + y_i) &= \left[\sum_{i=0}^{m-1}(x_i + y_i)\right] + (x_m + y_m)\\ &= \left[\sum_{i=0}^{m-1} x_i + \sum_{i=0}^{m-1} y_i\right] + (x_m + y_m)\\ &= \left[\left(\sum_{i=0}^{m-1} x_i\right) + x_m\right] + \left[\left(\sum_{i=0}^{m-1} y_i\right) + y_m\right]\\ &= \left[\sum_{i=0}^{m} x_i\right] + \left[\sum_{i=0}^{m} y_i\right],\end{aligned}$$

contradicting $m \in S$. Thus S is empty; equality holds for all $n \in \mathbb{Z}^{\geq 0}$. $\square$

Lemma C.11 (Uniqueness of Finite Cardinality). For $n, m \in \mathbb{Z}^{\geq 0}$, If there exists a bijection

$$f : [n] \to [m],$$

then $m = n$.

Proof. Let

$$S = \left\{n \in \mathbb{Z}^{\geq 0} : \text{there exist } m \in \mathbb{Z}^{\geq 0} \text{ and a bijection } f : [n] \to [m] \text{ with } n \neq m\right\}.$$

Assume for contradiction that S is nonempty, so by WOP it has a least element n. Let $m \in \mathbb{Z}^{\geq 0}$ and

$$f : [n] \to [m]$$

be a bijection with $n \neq m$.

We know $n \neq 0$, because otherwise $[n] = \emptyset$. Since f is surjective, this would imply $[m] = \emptyset$, and hence $m = 0$, contradicting $n \neq m$.

Thus $n > 0$, so by Lemma 3.18, $n \geq 1$. Since $n - 1 < n$, the minimality of n implies $n - 1 \notin S$, so the identity must hold for $n - 1$.

Also, since $n \geq 1$, $0 \in [n]$, so $f(0) \in [m]$. Hence $[m] \neq \emptyset$, which means $m \neq 0$. Since $m \in \mathbb{Z}^{\geq 0}$, we therefore have $m > 0$, and so by Lemma 3.18, $m \geq 1$.

To complete the WOP proof, we want to show that we can reduce the bijection $f : [n] \to [m]$ to the bijection $g : [n-1] \to [m-1]$. Since the identity is true for $n - 1$, this would imply that $n - 1 = m - 1$, yielding $n = m$. Since f is bijective, there exists a unique $x \in [n]$ such that

$$f(x) = m - 1.$$

Now, we have two cases. If $x = n - 1$, we let $g = f$. Otherwise, we will let $g(x) = f(n - 1)$. In all cases, we want $g(n - 1) = m - 1$ so we can simply restrict g to $[n - 1]$ by dropping this pair. Thus, in this second case we define $g : [n] \to [m]$ by

$$g(a) = \begin{cases} f(n-1), & a = x, \\ m - 1, & a = n - 1, \\ f(a), & \text{otherwise.} \end{cases}$$

Claim C.11.1 — In either case, g is a bijection.

Proof. If $x = n - 1$, then $g = f$, so bijectivity follows from f. Otherwise, if $x \neq n - 1$:

(Injective) Let $a, b \in [n]$ satisfy $g(a) = g(b)$.

If $a, b \neq x, n - 1$ then

$$f(a) = g(a) = g(b) = f(b),$$

so $a = b$ by injectivity of f.

Otherwise, if $a = x$, then $g(a) = f(n - 1)$.

1. If $b = x$, then $a = b$.

2. If $b = n - 1$, then $g(b) = m - 1 = f(x)$, so

$$f(n-1) = f(x),$$

contradicting the injectivity of f.

3. Finally, if $b \neq x, n-1$, then
$$f(n-1) = g(a) = g(b) = f(b),$$
so again we contradict the injectivity of f.

Hence $a = b$. The case $a = n-1$ follows analogously. Therefore g is injective.

(Surjective) Let $y \in [m]$ be arbitrary. Since f is surjective, there exists $a \in [n]$ such that $f(a) = y$. If $a = x$, then
$$y = f(x) = m - 1 = g(n-1).$$
If $a = n-1$, then
$$y = f(n-1) = g(x).$$
Otherwise,
$$y = f(a) = g(a).$$
Thus g is surjective. □

Hence g is a bijection with $g(n-1) = m-1$. Therefore, by Lemma 3.63, the restriction
$$g' : [n] \setminus \{n-1\} \to [m] \setminus \{m-1\}$$
is also a bijection. Since
$$[n] \setminus \{n-1\} = [n-1], \qquad [m] \setminus \{m-1\} = [m-1],$$
there exists a bijection
$$g' : [n-1] \to [m-1].$$
Thus $n-1 = m-1$, and so $n = m$, contradicting our choice of n and m. Thus S is empty, so the result holds in general. □

Lemma C.12 (Inverse of a Bijection). Let A and B be sets, and let $f : A \to B$ be a bijection. Define
$$f^{-1} = \{(b, a) \in B \times A : (a, b) \in f\}.$$
Then $f^{-1} : B \to A$ is a bijection.

Proof. We verify that f^{-1} is a function, injective, and surjective:

1. Function: Since f is surjective, for each $b \in B$ there exists $a \in A$ such that $(a, b) \in f$. Since f is injective, this a is unique. Therefore, by the definition of f^{-1}, for each $b \in B$ there exists a unique $a \in A$ such that $(b, a) \in f^{-1}$. Thus f^{-1} is a function.

2. Injective: Suppose $b_1, b_2 \in B$ satisfy
$$f^{-1}(b_1) = f^{-1}(b_2).$$
Let this common value be $a \in A$. Then
$$f(a) = b_1 \qquad \text{and} \qquad f(a) = b_2.$$
Since f is a function, $b_1 = b_2$. Thus f^{-1} is injective.

3. Surjective: Let $a \in A$ be arbitrary. Since f is a function, there exists $b \in B$ such that $f(a) = b$. By the definition of f^{-1},
$$f^{-1}(b) = a.$$
Thus f^{-1} is surjective.

Thus, $f^{-1} : B \to A$ is a bijection. □

Lemma C.13 (Cardinality of a Disjoint Union)**.** Let A and B be finite sets such that $A \cap B = \emptyset$. Then $|A \cup B| = |A| + |B|$.

Proof. Assume $|A| = m$ and $|B| = n$ for some $m, n \in \mathbb{Z}^{\geq 0}$, so that there exist bijections

$$f : [m] \to A, \qquad g : [n] \to B.$$

Now define $h : [m + n] \to A \cup B$ by

$$h(i) = \begin{cases} f(i), & 0 \leq i < m, \\ g(i - m), & m \leq i < m + n. \end{cases}$$

We claim that h is a bijection. (Injective) Suppose for $i, j \in [m + n]$, that $h(i) = h(j)$.

- Case 1: $i, j < m$. Then $f(i) = f(j)$, and since f is injective, $i = j$.
- Case 2: $i, j \geq m$. Then $g(i - m) = g(j - m)$, and since g is injective, $i - m = j - m$, so $i = j$.
- Case 3: $i < m, j \geq m$. Then $h(i) \in A, h(j) \in B$, so since $h(i) = h(j)$, $h(i) \in A \cap B$, which is a contradiction as the intersection is empty.
- Case 4: $i \geq m$ and $j < m$. This is symmetric to case 3; we would obtain a contradiction.

Thus, in all valid cases, $i = j$, so the function h is injective.

(Surjective) Let $x \in A \cup B$ be arbitrary. Then $x \in A$ or $x \in B$. If $x \in A$, then there exists $i \in [m]$ such that $f(i) = x$. Thus $h(i) = f(i) = x$, as desired. If $x \in B$, then there exists $j \in [n]$ such that $g(j) = x$. Since $0 \leq j < n$, $m \leq j + m < m + n$. Then $h(j + m) = g((j + m) - m) = g(j) = x$, as desired. Thus h is surjective.

Hence, $h : [m + n] \to A \cup B$ is a bijection. Therefore, by definition,

$$|A \cup B| = m + n.$$

Since $m = |A|$ and $n = |B|$, we obtain $|A \cup B| = |A| + |B|$. $\square$

Lemma C.14 (Subsets of Finite Sets Are Finite)**.** Let B be a finite set and let $A \subseteq B$. Then A is finite.

Proof. Let

$$S = \left\{ n \in \mathbb{Z}^{\geq 0} : \exists B \text{ with bijection } f : [n] \to B \text{ s.t. } \exists \text{non-finite } A \subseteq B \right\}.$$

Assume for contradiction that S is nonempty, so by WOP it has a least element n. Thus let there exist sets $A \subseteq B$ and a bijection

$$f : [n] \to B$$

such that A is not finite.

We first claim that $n \neq 0$. Otherwise, $f : [0] \to B$ is a bijection. Since $[0] = \emptyset$, we have $B = \emptyset$. Thus $A = \emptyset$, since $A \subseteq B$. But then A is finite, contradiction.

Hence $n > 0$, so by Lemma 3.18, $n \geq 1$, and so $n - 1 \in \mathbb{Z}^{\geq 0}$.

Since there exists a bijection $f : [n] \to B$, by definition B is finite. Since A is not finite, we have $A \neq B$. Thus by set theory there exists $b \in B \setminus A$.

Since f is bijective, there exists a unique $x \in [n]$ such that $f(x) = b$.

If $x = n - 1$, let $g = f$. Otherwise, define $g : [n] \to B$ by

$$g(a) = \begin{cases} f(n - 1), & a = x, \\ b, & a = n - 1, \\ f(a), & \text{otherwise.} \end{cases}$$

As in the proof of Lemma C.11, g is a bijection, with

$$g(n-1) = b.$$

By Lemma 3.63, the restriction

$$g' : [n] \setminus \{n-1\} \to B \setminus \{b\}$$

is a bijection. Since

$$[n] \setminus \{n-1\} = [n-1],$$

we have thus constructed a bijection

$$g' : [n-1] \to B \setminus \{b\}.$$

Since $b \notin A$ and $A \subseteq B$, by set theory

$$A \subseteq B \setminus \{b\}.$$

Thus, since A is not finite, $n-1 \in S$. But $n-1 < n$, contradicting the minimality of n.

Therefore S is empty, so every subset of a finite set is finite. □

Lemma C.15 (Cardinality of a Finite Partition)**.** Let B be a finite set. Let $\{A_i\}_{i \in \mathbb{Z}_{1,k}}$ for $k \in \mathbb{Z}^+$ be a partition of B. Then

$$\sum_{i=1}^{k} |A_i| = |B|.$$

Proof. Let

$$S = \{k \in \mathbb{Z}^+ : \exists \text{ finite set } B, \text{ partition } \{A_i\}_{i \in \mathbb{Z}_{1,k}}, \sum_{i=1}^{k} |A_i| \neq |B|\}.$$

Assume for contradiction S is nonempty. Then by WOP it has least element k. Thus let there exist finite set B and partition $\{A_i\}_{i \in \mathbb{Z}_{1,k}}$ such that $\sum_{i=1}^{k} |A_i| \neq |B|$. Since each $A_i \subseteq B$, Lemma 3.76 implies that each A_i is finite.

Claim C.15.1 — $k \neq 1$.

Proof. Assume for contradiction that $k = 1$. Then our partition has one set, so

$$\bigcup_{i \in \mathbb{Z}_{1,1}} A_i = A_1 = B.$$

Hence

$$\sum_{i=1}^{1} |A_i| = |A_1| = |B|,$$

contradiction. □

Hence, $k > 1$, so $k \geq 2$. Thus $k-1 \geq 1$, so since $k-1 < k$ and k is minimal in S, we have $k-1 \notin S$. Now let

$$B' = \bigcup_{i \in \mathbb{Z}_{1,k-1}} A_i.$$

Since $B' \subseteq B$, Lemma 3.76 implies that B' is finite.

Claim C.15.2 —

$$\{A_i\}_{i \in \mathbb{Z}_{1,k-1}}$$

is a partition of B'.

Proof. Recall Definition 3.78. The set satisfies (i) by the definition of B'. Moreover, because the original set of sets $\{A_i\}_{i \in \mathbb{Z}_{1,k}}$ satisfied pairwise disjointness, any subset of this is also pairwise disjoint, so this set also satisfies (ii). □

We know B' is finite and $\{A_i\}_{i \in \mathbb{Z}_{1,k-1}}$ is a partition of B'. Thus, since $k-1 \notin S$ but $k-1 \geq 1$, we know

$$\sum_{i=1}^{k-1} |A_i| = |B'|.$$

Claim C.15.3 — $B' \cap A_k = \emptyset$.

Proof. Suppose by contradiction there exists an $x \in B' \cap A_k$. Then $x \in B'$, so $x \in A_i$ for some $i \in \mathbb{Z}_{1,k-1}$. Hence $x \in A_i \cap A_k$, but this is a contradiction since these sets are disjoint. Thus $B' \cap A_k = \emptyset$. □

Claim C.15.4 — $B = B' \cup A_k$.

Proof. By definition,

$$B = \bigcup_{i \in \mathbb{Z}_{1,k}} A_i = \left[\bigcup_{i \in \mathbb{Z}_{1,k-1}} A_i \right] \cup A_k = B' \cup A_k.$$

□

Then by Lemma 3.75,

$$\begin{aligned} |B| &= |B' \cup A_k| \\ &= |B'| + |A_k| \\ &= \left(\sum_{i=1}^{k-1} |A_i| \right) + |A_k| \\ &= \sum_{i=1}^{k} |A_i|, \end{aligned}$$

contradicting the fact that $k \in S$. Hence S is empty, so the identity holds for all $k \in \mathbb{Z}^+$. □

Lemma C.16 (Scalar Distributivity)**.** Let R be a ring, let $r \in R$, and let $(a_i)_{i \in \mathbb{Z}^{\geq 0}}$ be a sequence in R. Then, for all $m \in \mathbb{Z}^{\geq 0}$,

$$r\left(\sum_{i=0}^{m} a_i\right) = \sum_{i=0}^{m} r a_i.$$

Proof. Let

$$S = \left\{ m \in \mathbb{Z}^{\geq 0} : r\left(\sum_{i=0}^{m} a_i\right) \neq \sum_{i=0}^{m} r a_i \right\}.$$

Assume for contradiction that S is nonempty, so by WOP it has a least element m.

We first have $m \neq 0$, since by the definition of summation,

$$r\left(\sum_{i=0}^{0} a_i\right) = r a_0 = \sum_{i=0}^{0} r a_i.$$

Thus $m > 0$, so by Lemma 3.18, $m \geq 1$. Hence $m - 1 \in \mathbb{Z}^{\geq 0}$ and $m - 1 < m$. By minimality of m, the statement holds for $m - 1$:

$$r\left(\sum_{i=0}^{m-1} a_i\right) = \sum_{i=0}^{m-1} r a_i.$$

Then by the definition of summation and distributivity,

$$\begin{aligned} r\left(\sum_{i=0}^{m} a_i\right) &= r\left(\sum_{i=0}^{m-1} a_i + a_m\right) \\ &= r\left(\sum_{i=0}^{m-1} a_i\right) + r a_m \\ &= \sum_{i=0}^{m-1} r a_i + r a_m \\ &= \sum_{i=0}^{m} r a_i. \end{aligned}$$

This contradicts $m \in S$. Therefore S is empty, so the lemma is true in general. □

D Polynomial Proofs

Lemma D.1 (Well-Definedness of Polynomial Addition)**.** For all $f, g \in R[x]$, we have

$$f + g \in R[x].$$

Thus polynomial addition is well-defined.

Proof. Since $f \in R[x]$, there exists $N \in \mathbb{Z}^{\geq 0}$ such that, for every $i \in \mathbb{Z}^{\geq 0}$,

$$i > N \implies a_i = 0.$$

Similarly, since $g \in R[x]$, there exists $M \in \mathbb{Z}^{\geq 0}$ such that, for every $i \in \mathbb{Z}^{\geq 0}$,

$$i > M \implies b_i = 0.$$

Now let

$$K = \max(N, M).$$

Then $K \geq N, M$, so $K \in \mathbb{Z}^{\geq 0}$ by transitivity. Let $i \in \mathbb{Z}^{\geq 0}$ and suppose that $i > K$. Since $N \leq K$ and $M \leq K$, we have

$$i > N, \qquad i > M.$$

Therefore,

$$a_i = 0, \qquad b_i = 0,$$

and hence $a_i + b_i = 0 + 0 = 0$. Thus $f + g \in R[x]$, so addition is well-defined. $\square$

Lemma D.2 (Well-Definedness of Polynomial Multiplication). For all $f, g \in R[x]$, we have

$$fg \in R[x].$$

Thus polynomial multiplication is well-defined.

Proof. Since $f \in R[x]$, there exists $N \in \mathbb{Z}^{\geq 0}$ such that, for every $i \in \mathbb{Z}^{\geq 0}$,

$$i > N \implies a_i = 0.$$

Similarly, since $g \in R[x]$, there exists $M \in \mathbb{Z}^{\geq 0}$ such that, for every $i \in \mathbb{Z}^{\geq 0}$,

$$i > M \implies b_i = 0.$$

Now let

$$K = N + M,$$

and let $n > N + M$. Then notice that in the sum

$$c_n = \sum_{i=0}^{n} a_i b_{n-i},$$

we have two cases: either $i > N$ or $i \leq N$. In the first case,

$$a_i = 0.$$

In the second case, we claim that

$$n - i > M.$$

First, since $n > N + M$, by Lemma 3.16, we have $n + (-i) > (N + M) + (-i)$, so

$$n - i > (N + M) + (-i).$$

Since $i \leq N$, either $i = N$ or $i < N$. If $i = N$ then $(N+M)+(-N) = (M+N)+(-N) = M+(N+(-N)) = M + 0 = M$. Thus $n - i > M$. If however, $i < N$, then there exists $v \in P$ such that $i + v = N$. Then $(-i) + (i + v) = (-i) + N$, so

$$((-i) + i) + v = (-i) + N$$

$$0 + v = (-i) + N$$

$$v = (-i) + N$$

$$v + (-N) = ((-i) + N) + (-N)$$

$$v + (-N) = (-i) + (N + (-N))$$

$$v + (-N) = (-i) + (0)$$

$$v + (-N) = -i$$
$$(-N) + v = -i.$$

Thus $-N < -i$. Hence,

$$(N + M) + (-N) < (N + M) + (-i)$$

by Lemma 3.16. By transitivity,

$$(N + M) + (-N) < (N + M) + (-i) < n - i$$
$$M < n - i$$
$$n - i > M,$$

as desired. Thus in this case $b_{n-i} = 0$. Therefore at every valid i, one of $a_i, b_{n-i} = 0$, so $a_i b_{n-i} = 0$. Thus $c_n = \sum_{i=0}^{n} 0 = 0$. Thus $fg \in R[x]$, so multiplication is well-defined. □

Lemma D.3 (Polynomial Ring)**.** $R[x]$ is a ring.

Proof. Let $f, g, h \in R[x]$ be defined by

$$f = (a_i)_{i \in \mathbb{Z}^{\geq 0}}, \qquad g = (b_i)_{i \in \mathbb{Z}^{\geq 0}}, \qquad h = (c_i)_{i \in \mathbb{Z}^{\geq 0}}.$$

We verify each of the ring axioms.

1. Commutativity $(+)$:

$$\begin{aligned} f + g &= (a_i + b_i)_{i \in \mathbb{Z}^{\geq 0}} \\ &= (b_i + a_i)_{i \in \mathbb{Z}^{\geq 0}} \\ &= g + f. \end{aligned}$$

2. Associativity $(+)$:

$$\begin{aligned} (f + g) + h &= ((a_i + b_i) + (c_i))_{i \in \mathbb{Z}^{\geq 0}} \\ &= (a_i + (b_i + c_i))_{i \in \mathbb{Z}^{\geq 0}} \\ &= f + (g + h). \end{aligned}$$

3. Commutativity $(\cdot)$: Let $fg = (m_i)_{i \in \mathbb{Z}^{\geq 0}}$, where

$$m_n = \sum_{i=0}^{n} a_i b_{n-i}.$$

Then we have

$$\sum_{i=0}^{n} a_i b_{n-i} = \sum_{i=0}^{n} b_{n-i} a_i.$$

Also, let $gf = (r_i)_{i \in \mathbb{Z}^{\geq 0}}$, where

$$r_n = \sum_{i=0}^{n} b_i a_{n-i}.$$

Now, define bijection $\phi : [n+1] \to [n+1]$ by $\phi(i) = n - i$.

We prove that ϕ is a bijection from $[n+1]$ to itself.

First, we prove that ϕ is well-defined. Let $i \in [n+1]$ be arbitrary. Then $0 \leq i \leq n$. ϕ outputs $n - i$. Thus, $0 \leq n - i \leq n$. So the output is in the set, and ϕ is well-defined.

Injective: Suppose $\phi(i) = \phi(j)$ so $n - i = n - j$. This means $-i = -j$, $i = j$. Therefore, ϕ is injective.

Surjective: Let x be any target value in the set and let the input $i = n - x$. Since $0 \leq x \leq n$, $0 \leq n - x \leq n$. So $n - x \in [n+1]$. Plugging in, $\phi(i) = n - (n - x) = x$. Thus, every element in the set can be obtained, and ϕ is surjective.

Thus ϕ is a bijection, so

$$\begin{aligned} m_n &= \sum_{i=0}^{n} b_{n-(n-i)} a_{n-i}. \\ &= \sum_{i=0}^{n} b_i a_{n-i} \\ &= r_n. \end{aligned}$$

Thus $fg = gf$.

4. Associativity $(\cdot)$:

   Let

   $$(fg)h = (m_n)_{n \in \mathbb{Z}^{\geq 0}}, \qquad f(gh) = (r_i)_{i \in \mathbb{Z}^{\geq 0}}.$$

   By definition,

   $$m_n = \sum_{j=0}^{n} \left(\sum_{i=0}^{j} a_i b_{j-i} \right) c_{n-j}, \qquad r_n = \sum_{i=0}^{n} a_i \left(\sum_{j=0}^{n-i} b_j c_{n-i-j} \right).$$

   We will show that these two expressions are always equal.

   By commutativity in R,

   $$m_n = \sum_{j=0}^{n} \left(\sum_{i=0}^{j} a_i b_{j-i} \right) c_{n-j} = \sum_{j=0}^{n} c_{n-j} \left(\sum_{i=0}^{j} a_i b_{j-i} \right).$$

   By Lemma C.16, as c_{n-j} does not depend on the value of i, this becomes

   $$\sum_{j=0}^{n} \left(\sum_{i=0}^{j} c_{n-j} (a_i b_{j-i}) \right).$$

   Then by associativity and commutativity,

   $$m_n = \sum_{j=0}^{n} \left(\sum_{i=0}^{j} a_i (b_{j-i} c_{n-j}) \right).$$

Claim D.3.1 — For any elements $x_{i,j} \in R$ indexed by $0 \le i \le j \le n$,

$$\sum_{j=0}^{n}\sum_{i=0}^{j} x_{i,j} = \sum_{i=0}^{n}\sum_{j=i}^{n} x_{i,j}.$$

Proof. Let

$$S = \left\{ n \in \mathbb{Z}^{\ge 0} : \sum_{j=0}^{n}\sum_{i=0}^{j} x_{i,j} \ne \sum_{i=0}^{n}\sum_{j=i}^{n} x_{i,j} \text{ for some } x_{i,j} \in R \right\}.$$

Assume for the sake of contradiction that S is nonempty. By WOP, let S have least element $n \ge 0$. Now if $n = 0$, we know

$$\sum_{j=0}^{0}\sum_{i=0}^{j} x_{i,j} = x_{0,0} = \sum_{i=0}^{0}\sum_{j=i}^{0} x_{i,j},$$

contradicting $0 \in S$. Hence $n > 0$, and by Lemma 3.18, $n \ge 1$, so $n - 1 \ge 0$.
But by the minimality of n in S, $n - 1 \notin S$, so the identity must hold for $n - 1$. Thus

$$\sum_{j=0}^{n-1}\sum_{i=0}^{j} x_{i,j} = \sum_{i=0}^{n-1}\sum_{j=i}^{n-1} x_{i,j}.$$

Now by definition,

$$\sum_{j=0}^{n}\sum_{i=0}^{j} x_{i,j} = \left(\sum_{j=0}^{n-1}\sum_{i=0}^{j} x_{i,j}\right) + \sum_{i=0}^{n} x_{i,n}$$

and

$$\sum_{i=0}^{n} x_{i,n} = \left(\sum_{i=0}^{n-1} x_{i,n}\right) + x_{n,n}.$$

Thus,

$$\begin{aligned}\sum_{j=0}^{n}\sum_{i=0}^{j} x_{i,j} &= \left(\sum_{j=0}^{n-1}\sum_{i=0}^{j} x_{i,j}\right) + \sum_{i=0}^{n} x_{i,n} \\ &= \left(\sum_{i=0}^{n-1}\sum_{j=i}^{n-1} x_{i,j}\right) + \left(\sum_{i=0}^{n-1} x_{i,n}\right) + x_{n,n}.\end{aligned}$$

By Lemma 3.68,

$$\begin{aligned}\left(\sum_{i=0}^{n-1}\sum_{j=i}^{n-1} x_{i,j}\right) + \left(\sum_{i=0}^{n-1} x_{i,n}\right) &= \sum_{i=0}^{n-1}\left[\left(\sum_{j=i}^{n-1} x_{i,j}\right) + x_{i,n}\right] \\ &= \sum_{i=0}^{n-1}\sum_{j=i}^{n} x_{i,j}.\end{aligned}$$

Hence,

$$\begin{aligned}\sum_{j=0}^{n}\sum_{i=0}^{j} x_{i,j} &= \left(\sum_{j=0}^{n-1}\sum_{i=0}^{j} x_{i,j}\right) + \sum_{i=0}^{n} x_{i,n} \\ &= \sum_{i=0}^{n-1}\sum_{j=i}^{n} x_{i,j} + x_{n,n} \\ &= \sum_{i=0}^{n}\sum_{j=i}^{n} x_{i,j}.\end{aligned}$$

But this contradicts $n \in S$. Hence S is empty, proving the claim. $\square$

Claim D.3.2 — For every $N \in \mathbb{Z}^{\geq 0}$, every $i \in \mathbb{Z}^{\geq 0}$, and sequences $(b_k), (c_k)$ in R, we claim

$$\sum_{j=i}^{i+N} b_{j-i}c_{(i+N)-j} = \sum_{k=0}^{N} b_k c_{N-k}.$$

Proof. Let

$$S = \left\{ N \in \mathbb{Z}^{\geq 0} : \exists i \in \mathbb{Z}^{\geq 0}, (b_k), (c_k) \ s.t. \ \sum_{j=i}^{i+N} b_{j-i}c_{(i+N)-j} \neq \sum_{k=0}^{N} b_k c_{N-k} \right\}.$$

Assume for contradiction that S is nonempty. By WOP, let m be its least element.

If $m = 0$, then for every $i, (b_k), (c_k)$,

$$\begin{aligned}\sum_{j=i}^{i+0} b_{j-i}c_{(i+0)-j} &= b_{i-i}c_{i-i} \\ &= b_0 c_0 \\ &= \sum_{k=0}^{0} b_k c_{0-k},\end{aligned}$$

contradicting $0 \in S$. Hence $m \neq 0$, so $m > 0$. Then by Lemma 3.18, we have $m \geq 1$, so $m - 1 \geq 0$. Since $m \in S$, choose i and sequences $(b_k), (c_k)$ for which the identity fails at m.

Now define new sequence $(d_k)_{k \in \mathbb{Z}^{\geq 0}}$ by

$$d_k = c_{k+1}.$$

Since m is least in $S, m - 1 \notin S$, so the identity holds for $m - 1$ for every pair of sequences, in particular for (b_k) and (d_k). It follows that

$$\sum_{j=i}^{i+(m-1)} b_{j-i}d_{(i+(m-1))-j} = \sum_{k=0}^{m-1} b_k d_{(m-1)-k}.$$

Note that for all $a, b \in R$,

$$\begin{aligned}\big((a-1)-b\big)+1 &= \big(a + ((-1)-b)\big)+1 \\ &= \big(a + (-b + (-1))\big)+1 \\ &= \big(a + (-b)\big) + ((-1)+1) \\ &= a - b.\end{aligned}$$

Thus by definition of d,

$$d_{(i+(m-1))-j} = c_{\big((i+(m-1))-j\big)+1} = c_{(i+m)-j},$$

and

$$d_{(m-1)-k} = c_{((m-1)-k)+1} = c_{m-k}.$$

Thus

$$\sum_{j=i}^{i+(m-1)} b_{j-i}c_{(i+m)-j} = \sum_{k=0}^{m-1} b_k c_{m-k}.$$

Now by definition of summation,

$$\begin{aligned}
\sum_{j=i}^{i+m} b_{j-i}c_{(i+m)-j} &= \left(\sum_{j=i}^{i+(m-1)} b_{j-i}c_{(i+m)-j}\right) + b_{(i+m)-i}c_{i+m-(i+m)} \\
&= \left(\sum_{k=0}^{m-1} b_k c_{m-k}\right) + b_m c_0 \\
&= \left(\sum_{k=0}^{m-1} b_k c_{m-k}\right) + b_m c_{m-m} \\
&= \sum_{k=0}^{m} b_k c_{m-k}.
\end{aligned}$$

Therefore the identity also holds at m, contradicting $m \in S$. Hence S is empty, so the claim is proven. □

Finally, we have

$$\begin{aligned}
m_n &= \sum_{j=0}^{n}\sum_{i=0}^{j} a_i(b_{j-i}c_{n-j}) \\
&= \sum_{i=0}^{n}\sum_{j=i}^{n} a_i(b_{j-i}c_{n-j}) && \text{by Claim D.3.1} \\
&= \sum_{i=0}^{n}\left(a_i \sum_{j=i}^{n}(b_{j-i}c_{n-j})\right) && \text{by Lemma C.16.}
\end{aligned}$$

Now, let $N = n - i$. Indeed, since $0 \le i \le n$, we have $n - i \ge 0$, so we can apply Claim D.3.2 to obtain

$$\begin{aligned}
\sum_{j=i}^{i+(n-i)} b_{j-i}c_{(i+(n-i))-j} &= \sum_{k=0}^{n-i} b_k c_{(n-i)-k} \\
\sum_{j=i}^{n} b_{j-i}c_{n-j} &= \sum_{k=0}^{n-i} b_k c_{(n-i)-k}.
\end{aligned}$$

Hence,

$$\begin{aligned}
m_n &= \sum_{i=0}^{n}\left(a_i \sum_{k=0}^{n-i} b_k c_{(n-i)-k}\right) && \text{by Claim D.3.2} \\
&= r_n,
\end{aligned}$$

for every integer $n \geq 0$. Thus

$$(fg)h = f(gh),$$

and we are done.

5. Distributivity: Let $f(g+h) = (m_i)_{i\in\mathbb{Z}^{\geq 0}}$. Then

$$g + h = (b_i + c_i)_{i\in\mathbb{Z}^{\geq 0}},$$

so we have

$$m_n = \sum_{i=0}^{n} a_i(b_{n-i} + c_{n-i}).$$

Then by Lemma 3.68,

$$\begin{aligned} m_n &= \sum_{i=0}^{n} (a_i b_{n-i} + a_i c_{n-i}) \\ &= \sum_{i=0}^{n} a_i b_{n-i} + \sum_{i=0}^{n} a_i c_{n-i} \\ &= (fg)_n + (fh)_n. \end{aligned}$$

Hence $f(g+h) = fg + fh$.

6. Zero: Let

$$0 = (0)_{i\in\mathbb{Z}^{\geq 0}}.$$

Then

$$\begin{aligned} f + 0 &= (a_i + 0)_{i\in\mathbb{Z}^{\geq 0}} \\ &= (a_i)_{i\in\mathbb{Z}^{\geq 0}} \\ &= f. \end{aligned}$$

7. Negatives: Let

$$-f = (-a_i)_{i\in\mathbb{Z}^{\geq 0}}.$$

Then

$$\begin{aligned} f + (-f) &= (a_i + (-a_i))_{i\in\mathbb{Z}^{\geq 0}} \\ &= (0)_{i\in\mathbb{Z}^{\geq 0}} \\ &= 0. \end{aligned}$$

8. One: Let

$$1 = (e_i)_{i\in\mathbb{Z}^{\geq 0}}$$

such that

$$e_i = \begin{cases} 1 & i = 0 \\ 0 & i > 0. \end{cases}$$

Then by commutativity of multiplication (proved previously), $g \cdot 1 = 1 \cdot g$. Let this common value be

$$(m_i)_{i\in\mathbb{Z}^{\geq 0}}.$$

Thus we have

$$m_n = \sum_{i=0}^{n} e_i b_{n-i}.$$

Now if $n = 0$, we have
$$m_0 = e_0 b_0 = b_0.$$
Otherwise, $n > 0$, so
$$\begin{aligned} m_n &= e_0 b_n + \sum_{i=1}^{n} e_i b_{n-i} \\ &= 1 \cdot b_n + \sum_{i=1}^{n} 0 \cdot b_{n-i} \\ &= b_n + \sum_{i=1}^{n} 0 \\ &= b_n + 0 \\ &= b_n. \end{aligned}$$
Since $m_i = b_i$ for all $i \in \mathbb{Z}^{\geq 0}$, we have
$$g \cdot 1 = (m_i)_{i \in \mathbb{Z}^{\geq 0}} = (b_i)_{i \in \mathbb{Z}^{\geq 0}} = g.$$
□

Lemma D.4 (Evaluation Homomorphism). Evaluation of polynomials $f, g \in R[x]$ at $r \in R$ respects addition and multiplication. That is,
$$f(r) + g(r) = (f+g)(r), \qquad f(r)g(r) = (f \cdot g)(r).$$

Proof. First, we show that evaluation respects addition.

If $f = 0$, then
$$(f+g)(r) = g(r) = 0 + g(r) = f(r) + g(r),$$
and analogously for $g = 0$.

Now if $f, g \neq 0$, let
$$\deg(f) = n, \qquad \deg(g) = m,$$
and write
$$f = (f_i)_{i \in \mathbb{Z}^{\geq 0}}, \qquad g = (g_i)_{i \in \mathbb{Z}^{\geq 0}}.$$
Since $m + n \geq n, m$, all $(f_i), (g_i)$ for $i > m + n$ are zero. Hence, by definition,
$$\begin{aligned} (f+g)(r) &= \sum_{i=0}^{m+n} (f_i + g_i) r^i \\ &= \sum_{i=0}^{m+n} (f_i r^i + g_i r^i) \\ &= \sum_{i=0}^{m+n} f_i r^i + \sum_{i=0}^{m+n} g_i r^i && \text{by Lemma 3.68} \\ &= \sum_{i=0}^{n} f_i r^i + \sum_{i=0}^{m} g_i r^i \\ &= f(r) + g(r). \end{aligned}$$
Thus evaluation respects addition.

We now show that evaluation respects multiplication. If $f = 0$ or $g = 0$, then

$$(f \cdot g)(r) = 0 = f(r)g(r).$$

Thus, suppose $f, g \neq 0$, and let

$$\deg(f) = n, \qquad \deg(g) = m.$$

Write

$$f = (f_i)_{i \in \mathbb{Z}^{\geq 0}}, \qquad g = (g_i)_{i \in \mathbb{Z}^{\geq 0}}.$$

By Lemma C.16,

$$\begin{aligned}(f \cdot g)(r) &= \sum_{k=0}^{m+n} \left(\sum_{i=0}^{k} f_i g_{k-i} \right) r^k \\ &= \sum_{k=0}^{m+n} \sum_{i=0}^{k} f_i g_{k-i} r^k.\end{aligned}$$

Then by Claim D.3.1,

$$(f \cdot g)(r) = \sum_{i=0}^{m+n} \sum_{k=i}^{m+n} f_i g_{k-i} r^k.$$

Now let $j = k - i$. Then $k = i + j$, so by Claim D.3.2 with $N = m + n - i, b_k = g_k$, and $c_k = r^{m+n-k}$, we obtain

$$\sum_{k=i}^{m+n} g_{k-i} r^k = \sum_{j=0}^{m+n-i} g_j r^{i+j}.$$

Thus,

$$f_i \sum_{k=i}^{m+n} g_{k-i} r^k = f_i \sum_{j=0}^{m+n-i} g_j r^{i+j},$$

so by Lemma C.16,

$$\sum_{k=i}^{m+n} f_i g_{k-i} r^k = \sum_{j=0}^{m+n-i} f_i g_j r^{i+j},$$

and so,

$$(f \cdot g)(r) = \sum_{i=0}^{m+n} \sum_{j=0}^{m+n-i} f_i g_j r^{i+j}.$$

Since $f_i = 0$ for all $i > n$, and $g_j = 0$ for all $j > m$, all terms beyond the indices $0 \leq i \leq n$ and $0 \leq j \leq m$ are equal to 0. Hence,

$$(f \cdot g)(r) = \sum_{i=0}^{n} \sum_{j=0}^{m} f_i g_j r^{i+j}.$$

By Lemma 3.33, $r^{i+j} = r^i r^j$, so

$$\begin{aligned}
(f \cdot g)(r) &= \sum_{i=0}^{n} \sum_{j=0}^{m} f_i g_j r^i r^j \\
&= \sum_{i=0}^{n} \sum_{j=0}^{m} f_i r^i g_j r^j \\
&= \sum_{i=0}^{n} \left[f_i r^i \left(\sum_{j=0}^{m} g_j r^j \right) \right] && \text{by Lemma C.16} \\
&= \left(\sum_{i=0}^{n} f_i r^i \right) \left(\sum_{j=0}^{m} g_j r^j \right) && \text{by Lemma C.16} \\
&= f(r) g(r).
\end{aligned}$$

Thus evaluation respects multiplication. □